INCH-POUND

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SUPERSEDING
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(SEE 6.9)

MILITARY SPECIFICATION

AIR CONDITIONERS: VERTICAL AND HORIZONTAL, COMPACT, MULTIPLE POWER INPUT

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers compact, vertical and horizontal air conditioners capable of operating from multiple power inputs.

1.2 Classification. The air conditioners should be of the following types, sizes as specified (see 6.2 and 6.10):

Type I - Vertical	
Size	Power Input
C - 18,000 Btu/hr	- 208 V, 3-phase, 50/60/400 Hz.
Type II - Horizontal	
Size	Power Input
B - 18,000 Btu/hr	- 230 V, single-phase, 50/60 Hz.
	- 208 V, 3-phase, 50/60/400 Hz.

SPECIFICATIONS

FEDERAL

BB-F-1421	- Fluorocarbon Refrigerants.
NN-P-71	- Pallet, Material Handling, Wood, Stringer Construction, 2-Way and 4-Way (Partial).
QQ-S-781	- Strapping, Steel, and Seals.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Belvoir Research, Development, and Engineering Center, ATTN: STRBE-TSE, Fort Belvoir, VA 22060-5606 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 4120

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-636	- Boxes, Shipping, Fiberboard.
PPP-C-843	- Cushioning Material, Cellulosic.
PPP-S-30	- Sacks, Shipping, Paper (Cushioned).

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MIL-P-116	- Preservation, Methods of.
MIL-V-173	- Varnish, Moisture and Fungus Resistant (for Treatment of Communications, Electronic, and Associated Equipment).
MIL-P-514	- Plates, Identification, Instruction and Marking, Blank.
MIL-T-704	- Treatment and Painting of Materiel.
MIL-B-7883	- Brazing of Steels, Copper, Copper Alloys, Nickel Alloys, Aluminum and Aluminum Alloys.
MIL-P-15024	- Plate, Tags and Bands for Identification of Equipment.
MIL-P-15024/6	- Plate, Identification.
MIL-C-46168	- Coating, Aliphatic Polyurethane, Chemical Agent Resistant.

STANDARDS

FEDERAL

FED-STD-313	- Material Safety Data Sheets, Preparation and Submission of.

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-454	- Standard General Requirements for Electronic Equipment.
MIL-STD-461	- Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference.

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MIL-STD-462	- Electromagnetic Interference Characteristics, Measurement of.
MIL-STD-781	- Reliability Testing for Engineering Development, Qualification and Production.
MIL-STD-810[1/]	- Environmental Test Methods and Engineering Guidelines (see 6.8).
MIL-STD-882	- System Safety Program Requirements.
MIL-STD-889	- Dissimilar Metals.
MIL-STD-1190	- Minimum Guidelines for Level C Preservation, Packing and Marking.
MIL-STD-1472	- Human Engineering Design Criteria for Military Systems, Equipment and Facilities.
MIL-STD-1474	- Noise Limits for Army Materiel.

HANDBOOKS

MILITARY

MIL-HDBK-781	- Reliability Test Methods, Plans, and Environments for Engineering Development, Qualification, and Production.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN; NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.1.2 Government drawings. The following other Government drawings form a part of this specification to the extent specified herein. Unless otherwise specified, the issues are those in effect on the date of the solicitation.

DRAWINGS

ME

TA13229E4200	- Air Conditioner, Horizontal, Compact, 18,000 Btu/hr, 230 Volt-AC, Single-Phase, 50/60 Hertz, and 208 Volt-AC, 3 Phase, 50/60/400 Hertz.
TA13229E4250	- Air Conditioner, Vertical, Compact, 18,000 Btu/hr, 208 Volt-AC, 3-Phase, 50/60/400 Hertz.
TA13222E9905	- Blast Damper Assembly.
13222E9904	- Cover, Fresh Air.
13226E1362	- Cover, Air Conditioner.

(Copies of drawings, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer).

1/ The revision C to MIL-STD-810 should be used (see 6.8).

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2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

ACOUSTICAL SOCIETY OF AMERICA (ASA)

ASA - 53 - Preferred Frequencies, Frequency Levels, and Band Numbers for Acoustical Measurements.

(Applications for copies should be addressed to the Acoustical Society of America, 335 East 45th Street, New York, N.Y. 10017.)

AIR MOVING AND CONDITIONING ASSOCIATION (AMCA)

AMCA STD 210 - Laboratory Methods of Testing Fans for Ratings.

(Application for copies should be addressed to the Air Moving and Conditioning Association, 205 West Touhy Avenue, Park Ridge, IL 60068).

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI S1.1 - Acoustical Terminology (Including Mechanical Shock and Vibration).

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10018.)

AMERICAN SOCIETY OF HEATING, REFRIGERATING, AND AIR CONDITIONING ENGINEERS, INC. (ASHRAE)

ASHRAE STD 16 - Methods of Testing for Rating Room Air Conditioners.
ASHRAE STD 37 - Methods of Testing for Rating Unitary Air Conditioning and Heat Pump Equipment.
ASHRAE STD 51 - Laboratory Methods of Testing Fans for Rating.

(Application for copies should be addressed to ASHRAE Publication Sales, 1791 Tullie Circle N.E., Atlanta, GA. 30329).

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

Boiler and Pressure Vessel Code, Section IX, Welding and Brazing Qualifications.

(Application for copies should be addressed to the American Society of Mechanical Engineers, 345 East 47th Street, New York, NY 10017.)

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AMERICAN WELDING SOCIETY (AWS)

D1.1 - Structural Welding Code, Steel.
D1.2 - Structural Welding Code, Aluminum.

(Application for copies should be addressed to the American Welding Society, 550 N.W. LeJeune Road, P. O. Box 351040, Miami, FL 33135.)

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

J429 - Mechanical and Material Requirements for Externally Threaded Fasteners.
J492 - Rivets and Riveting.

(Application for copies should be addressed to the Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale, PA 15096.)

SOUTH COAST AIR QUALITY MANAGEMENT CONTROL DISTRICT

Rule 442 - Usage of Solvents.

(Application for copies should be addressed to the South Coast Air Quality Management Control District, 9150 East Flair Drive, El Monte, CA 91731).

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein the text of this specification shall take precedence. Nothing in this specification, however, supercedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. The air conditioners shall be in accordance with the following top assemblies and as specified herein:

Type I - Vertical

Size C - 18,000 Btu/hr - TA13229E4250

Type II - Horizontal

Size B - 18,000 Btu/hr - TA13229E4200

The air conditioners shall provide cooling, heating and ventilation utilizing the electrical power supplies as specified herein (see tables I, II and III). The air conditioners shall be equipped with a motor controller and logic section as specified (see 3.12.2 and 3.12.3). When specified (see 6.2) accessories shall be provided with each air conditioner as specified in 3.17.

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3.1.1 Drawings. The drawings forming a part of this specification are end product drawings. No deviation from the prescribed dimensions or tolerances is permissible without prior approval of the contracting officer. Where tolerances could cumulatively result in incorrect fits, the contractor shall provide tolerances within those prescribed on the drawings to ensure correct fit, assembly, and operation of the air conditioners. Any data (e.g., shop drawings, layouts, flow sheets, and processing procedures) prepared by the contractor or obtained from a vendor to support fabrication and manufacture of the production item shall be made available, upon request, for inspection by the contracting officer or designated representative.

3.2 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 4.3 and 6.3).

3.3 Reliability and maintainability.

3.3.1 Reliability. The mean-time-between-failures (MTBF) shall be as follows when tested as specified in 4.5.2.19:

Upper Test MTBF (θ_0) = 960 hrs
Lower Test MTBF (θ_1) = 480 hrs

3.3.2 Maintainability.

3.3.2.1 Maintenance ratio. The air conditioners shall have a maintenance ratio of not more than 0.03 when tested as specified in 4.5.2.21. The maintenance ratio is defined as the ratio of the total active maintenance manhours required (scheduled and unscheduled) to the total operating time. Manhours for repair of replaced components and scheduled before-and-after operation checks are excluded. A maintenance schedule shall be furnished prior to the start of any testing. Not more than 25 percent of repairs shall require maintenance at the general support level.

3.3.2.2 Scheduled maintenance. The air conditioners shall not require scheduled maintenance more often than every 250 hrs of operation when tested as specified in 4.5.2.21. Scheduled maintenance shall not be required at any level higher than organizational maintenance.

3.3.2.3 Fault indicators. LED indicators on the control panel shall provide identification of the following fault conditions:

a. Overcurrent (see 3.12.2).
b. Motor controller overheat (see 3.12.2).
c. Over or under voltage (see 3.12.2).
d. Temperature/pressure fault (see 3.9.3, 3.9.4, 3.10.4 and 3.10.5).

3.4 Materials. Material shall be as specified herein and as shown on the applicable drawings. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification.

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3.4.1 Material deterioration prevention and control. The air conditioners shall be fabricated from compatible materials, inherently corrosion resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable operating and storage environments to which the air conditioner may be exposed.

3.4.2 Dissimilar metals. Dissimilar metals shall not be used in intimate contact with each other unless protected against galvanic corrosion. Dissimilar metals and methods of protection are defined and detailed in MIL-STD-889.

3.4.3 Identification of materials and finishes. The contractor shall identify the specific material, material finish or treatment for use with components and subcomponents, and shall make information available upon request to the contracting officer or designated representative.

3.4.4 Recovered materials. For the purpose of this requirement, recovered materials are those materials that have been collected from solid waste and reprocessed to become a source of raw materials, as distinguished from virgin raw materials. The components, pieces and parts incorporated in the air conditioner may be newly fabricated from recovered materials to the maximum extent practicable, provided the air conditioners produced meets all other requirements of this specification. Used, rebuilt or remanufactured components, pieces and parts shall not be incorporated in the air conditioners.

3.5 Performance characteristics.

3.5.1 Cooling operating temperatures.

3.5.1.1 High temperature operation. After being subjected to not less than 120 °F for a period of 4 hours, the air conditioners shall start and operate in the cooling mode with air at not less than 120 °F ambient temperature entering the evaporator and condenser.

3.5.1.2 Low temperature operation.

3.5.1.2.1 Type I, size C. Type I, size C units shall operate in cooling mode without forming frost or ice on the evaporator at 55 °F ambient temperature with air at 70 ±2 °F dry bulb and 60 ±3 °F wet bulb entering the evaporator.

3.5.1.2.2 Type II, size B. After being subjected to 70 ±5 °F on the front and 0 ±2 °F on the rear for a period of 10 hours, the air conditioners shall start and operate in the cooling mode as specified with air at 70 ±5 °F entering the evaporator and air at 0 ±2 °F entering the condenser.

3.5.2 Cooling capacity. When operating at the specified conditions, the cooling capacities of the air conditioners shall be not less than as specified in table I.

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TABLE I. Cooling capacities.

Unit	Condition	Ambient temperature (dry bulb)	Evaporator inlet temperature		Minimum net sensible cooling capacity	Minimum net total cooling capacity
			dry bulb	wet bulb		
		°F	°F	°F	Btu/hr	Btu/hr
Type I						
Size C	A	120	90	75	12,100	18,500
	B	95	80	67	14,000	21,500
Type II						
Size B	A	120	90	75	14,000	18,500
	B	95	80	67	14,700	19,500

Note: Data shown for 0-inch water gage (WG) (free air) external static pressure on the evaporator for type I units and 0.25 inch WG for type II units.

3.5.3 Operating pressures. When the air conditioners are operating as specified in 3.5.1.1 or 3.5.2, the condensing and suction pressures shall be as specified in table II.

TABLE II. Condensing and suction pressures (psig).

Unit	Operating conditions (see 3.5.1.1)			Operating conditions (see 3.5.2)	
	Condensing pressure (max)	(min)	Suction pressure (min)	Condensing pressure (min)	Suction pressure (min)
Type I Size C	470	-	50	440	65
Type II Size B	470	160	50	425	50

3.5.4 Evaporator airflow. With the mode selector switch set on VENT, the evaporator airflow shall be not less than as specified in table III, column (1). The airflow shall be obtained with the external static pressure resistance (inch/wg) as specified in the note to table I.

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3.5.5 Heating.

3.5.5.1 Heating capacity. When operated in the heating mode, in an ambient temperature of 75 to 80 °F, the heating capacity shall be not less than as specified in table III, column (2).

3.5.5.2 Low temperature operation. After being subjected to -50 ±5 °F for a period of not less than 4 hours, the air conditioners shall start, operate and restart in the heating mode with air at -50 ±5 °F entering the evaporator.

3.5.6 Power consumption. When the air conditioners are operating at the conditions shown in 3.5.2 or 3.5.5.1, the total power consumption shall be not more than specified in table III, column (3) at a power factor not less than specified in table III, column (4).

TABLE III. Airflow, heating capacity and power consumption.

Unit	Evaporator Airflow (minimum SCFM) (1)	Heating (minimum Btu/hr) (2)	Power Consumption (maximum KW) (3)	Power Factor (minimum) (4)
Type I Size C	570	12,000	6.5Kw	0.9
Type II Size B	580	13,400	6.5Kw	0.9

3.5.7 Electrical power supply. The air conditioners shall operate as specified herein when supplied with external electrical power specified below:

Type I

Size C - 208 V, 3-phase, 50/60/400 Hz.

Type II

Size B - 230 V, single-phase, 50/60 Hz.
- 208 V, 3-phase, 50/60/400 Hz.

3.6 Human factors engineering. The air conditioners shall conform to human factors engineering design criteria as described in MIL-STD-1472. Special design emphasis shall be given, but not limited to: general requirements (4), control/display integration (5.1), visual displays (5.2), controls (5.4), labeling (5.5), design for maintainability (5.9) and hazards and safety (5.13) of MIL-STD-1472, as applicable.

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3.6.1 Noise. With microphones positioned as stated in 4.5.2.26 and with air conditioners operating on the cooling cycle, the noise levels of the air conditioners when tested in an anechoic chamber (free field), shall not exceed the values shown in table IV at the flat weighing network scale defined in ANSI S1.1. Octave band center frequencies shall be in accordance with ASA 53. Reference sound pressure shall be 0.0002 microbar. In addition, MIL-STD-1474, 5.1.2.5.1.2 and 5.1.3 shall apply.

TABLE IV. Corrected sound pressure levels in decibels.

Octave band center Frequencies - CPS	Evaporator side (db max)	Condenser side (db max)
Type I	Size C	
63	65	60
125	76	73
250	75	74
500	77	71
1000	69	71
2000	66	70
4000	62	69
8000	56	69
Type II	Size B	
63	60	60
125	66	64
250	69	63
500	68	65
1000	65	66
2000	60	66
4000	54	64
8000	65	55

3.7 Safety. All rotating or reciprocating parts and other parts subject to high operational temperatures that are of such nature or are so located as to be a hazard to operating personnel shall be guarded or insulated to the extent necessary to eliminate the hazard. Electrical equipment shall be effectively guarded and grounded to protect all persons and objects from electrical shock hazard in accordance with MIL-STD-454 requirement 1. Electrical potential from the air conditioner casing to the ground shall be not more than 1 volt (V). System safety design shall be in accordance with the criteria in MIL-STD-882.

3.8 Environmental. The air conditioners shall be capable of withstanding exposure to the following conditions:

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3.8.1 Storage conditions. The air conditioners shall withstand storage temperatures from -60 to +160 °F without impairment of specified capabilities.

3.8.2 Operating conditions. The air conditioners shall start and operate through a temperature range of -50 to +120 °F as specified herein.

3.8.3 Salt fog and humidity. When subjected to the tests specified in 4.5.2.15 and 4.5.2.16, no part of the air conditioners shall show evidence of deterioration, corrosion, or change in tolerance limits that could prevent the unit from meeting operational requirements. The unit shall show no evidence of peeling on paint, malfunctioning, refrigerant leaks, rough operation, electrical shorts, erratic pressure, or other irregular operation.

3.8.4 Rain. When subjected to the tests specified in 4.5.2.14, the air conditioners shall operate without functional damage or impairment of operation. There shall be no penetration of water from the evaporator section to the dry side of the enclosure or into or on any electrical component, in particular the motors, electrical connectors, terminals, switches, or any controls that could result in an electrical short, cause damage from rust or corrosion, or become a hazard to personnel.

3.8.5 Fungus and moisture resistance. The electrical circuitry, including all components and connections, except as specified herein, shall be protected from the effects of fungus growth and moisture by an overall treatment with a varnish conforming to MIL-V-173, composition as specified in 3.8.5.1, with 1-percent copper 8-quinolinolate (by weight) based on the nonvolatile content of the varnish. Components or circuit elements that are inherently fungus-and moisture-resistant or which are hermetically sealed need not be treated. Components or circuit elements whose functions will be adversely affected by the varnish coating shall not be treated. When used, the varnish shall be applied by spray, brush, or a combination of both to give a dry-film thickness of not less than 1 mil to component or element surfaces. Surfaces shall be cleaned and prepared to be free from all foreign matter that would interfere with adherence or function of the varnish.

3.8.5.1 Varnish composition. Composition I shall be used unless composition II must be used to comply with local air-pollution regulations in the application of the varnish. When composition II is used, the contractor shall provide evidence to the Government that the use of composition II is required, and shall certify that the composition II material complies with Southcoast Air Quality Management Control District Rule 442 - Usage of Solvents. The certification shall be made available for review by the contracting officer.

3.9 Refrigeration system. The refrigeration system shall be of the vapor-cycle type and furnished with a full charge of refrigerant conforming to BB-F-1421, type 22.

3.9.1 Moisture content. The moisture content in the system shall be not more than 25 parts per million (ppm) by weight in the liquid phase.

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3.9.2 Function. Refrigerant flow to the evaporator shall be controlled by the evaporator thermal expansion valve and compressor operation. When the predetermined temperature is attained within the conditioned space, the logic assembly shall deenergize the compressor relay, stopping the compressor and flow of refrigerant to the evaporator, causing deenergizing of the equalizing valve allowing equalization of the low and high side pressures.

3.9.3 High-pressure cutout switch. When the condensing pressure reaches the high pressure level specified in table V, the high pressure cutout switch shall open the 12 vdc control circuit causing shutdown of the compressor and illumination of the TEMPERATURE/PRESSURE fault light.

3.9.4 Low-pressure cutout switch. When the suction pressure decreases to the low pressure level specified in table V, the low pressure cutout switch shall open the 12 vdc control circuit causing shutdown of the compressor and illumination of the TEMPERATURE/PRESSURE fault light.

TABLE V. High and low cutout pressures.

Unit	Compressor high-pressure cutout (±10 psig)	Compressor low-pressure cutout (±5 psig)
Type I Size C	480	7
Type II Size B	480	7

3.9.5 Refrigerant system leakage. The entire refrigerant system shall withstand pressures up to 425 psig without deformation of any components or any detectable leakage when tested as specified in 4.5.2.3.1 and 4.5.2.3.2.

3.9.6 Material Safety Data Sheets. Material Safety Data Sheets prepared in accordance with FED-STD-313 shall be furnished for refrigerant (see 6.6.1).

3.10 Controls. Controls for automatic operation of the air conditioner shall be as specified in 3.10.1 through 3.10.5.

3.10.1 Control box. The mode selector switch, temperature selection control and fault indicators (see 3.3.2.3) shall be mounted in an integral control box in the front of the unit. The control box shall be easily removable for remote operation.

3.10.2 Mode selector switch. The mode selector switch shall control operating modes of the air condition of the air conditioner as follows:

a. With the switch in the OFF/RESET position, and with power connected to the air conditioner, the compressor crankcase heater shall be energized and the fan motor(s), heaters, and compressor shall be deenergized.

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b. With the switch in the VENT position, the evaporator fan motor shall start and operate. The compressor, condenser fan motor (if separate) and the heater elements shall be deenergized when the air conditioner is operating in the ventilation mode.
c. With the switch in the COOL position, the evaporator fan motor, condenser fan motor (if separate) and, the compressor motor shall start and operate. The electrical heating elements shall be deenergized when the air conditioner is operating in the cooling mode.
d. With the switch in the HEAT position, the electrical heating elements shall be energized and the evaporator fan motor shall operate. The compressor and condenser fan motor, if separate, shall not operate when the air conditioner is operating in the heating mode.

3.10.3 Conditioned-air thermostat. The conditioned-air thermostat shall control the cooling and heating temperatures automatically between 60 and 90 °F by regulating the operation of the compressor, fan motor(s) and the heater elements through the logic assembly.

3.10.4 Heater high-limit cutout switch. The heater high-limit cutout switch shall open the 12 vdc control circuit causing the logic assembly to deenergize the electrical heating elements to prevent overheating or damage to the air conditioner and illuminate the TEMPERATURE/PRESSURE fault light.

3.10.5 Coil frost switch. The coil frost switch shall open the 12 vdc control circuit at a sensed temperature of 30.5 ±1.5 °F causing the logic assembly to deenergize the compressor. The coil frost switch shall reset at a temperature of 45 ±2 °F, causing the logic assembly to energize the compressor.

3.11 Evaporator compartment. The air leakage from the evaporator compartment shall be not more than 20 standard cubic feet minute (SCFM) when the evaporator compartment is subjected to an internal pressure of 1.0-inch WG with the evaporator air discharge, return air, and condensate drains blocked, and when applicable, with a duct attached to the chemical-biological-radiological (CBR) opening for supply air. The fresh-air damper shall be closed if both a CBR opening and a fresh-air opening are present. If only a fresh-air opening is present, it shall remain open and the duct shall be attached to it.

3.12 Electrical system. The air conditioners shall be in accordance with the top assemblies specified in 3.1 and the following:

3.12.1 Wiring. Wiring shall be secured in compact harnesses and attached to frame members with insulated clamps at close intervals to ensure a neat and orderly cable run. Cables and wires shall not be spliced at any point throughout the length of their runs. Wiring shall not be attached to panels or other components that require removal during maintenance, except for actual electrical connections to components. Electrical isolators shall be used between all wiring and metal components to prevent abrasive action to wire insulation.

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3.12.2 Motor controller. The motor controller shall have the ability to:

a. Accept nominal input power of 208V, 3 phase, 50/60/400 Hz and 230V, single phase, 50-60 Hz.
b. At the power inputs listed in table VI, provide a "soft start" output to the air conditioner motors under all operating conditions specified herein. The soft start-up current to the motor controller shall not exceed the full load current during steady state operation.
c. Provide power to the external logic assembly (see 3.12.3).
d. Withstand voltage and frequency variations as specified without failure or tripping.
e. Interface with the external logic assembly by:

(1) Accepting start/run and stop/reset signals from the external logic assembly.
(2) Provide individual signals to the external logic assembly for the following faults:

(a) Over and under voltage in the input power source (see 3.12.4).
(b) Over current in output power.
(c) Over temperature of the motor controller that could damage components or prevent operation as specified.

f. Automatically shutdown in the event of over/under input power voltage, output overcurrent and over temperature of the motor controller.

3.12.3 Logic assembly. The logic assembly shall:

a. Accept signals from the mode selector switch and temperature selection control and control operation of the air conditioner.
b. Provide start/run and stop/reset signals to the motor controller.
c. Accept fault signals from the motor controller as specified in 3.12.2.e(2) and illuminate applicable fault lights.
d. Accept control circuit fault signals, including low and high pressure cutouts, motor thermal overloads, heater over temperature and evaporator coil frost switch cutout, and illuminate the applicable fault lights.

3.12.4 Variable voltage and frequency. The air conditioners shall be capable of rated operation utilizing power supplies with voltages and frequencies as specified in table VI.

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TABLE VI. Voltage and frequency variation.

Condition	Phase	Voltage	Frequency
1	3	219	420
2	3	197	50
3	3	197	420
4	3	219	50
5	1	242	63
6	1	242	50
7	1	218	63
8	1	218	50

Note 1. Conditions 5, 6, 7 and 8 apply to type II (horizontal) air conditioners only.
Note 2. Conditions 1, 2, 3 and 4 line currents shall not exceed 21 amperes. Conditions 5, 6, 7 and 8 line currents shall not exceed 35 amperes.

3.13 Vibration. When subjected to the test specified in 4.5.2.17, the air conditioners shall show no evidence of structural damage, misalignment or malfunction of components, leaks, fractures, abnormal vibration, or any other irregular operation.

3.14 Electromagnetic interference (EMI). The air conditioners shall comply with the requirements of MIL-STD-461, part 4, as specified herein when tested as specified in 4.5.2.20.

a. RE02 radiated emissions, electric field, 0.15 MHz to 1000 MHz, broadband and narrow band (figure 4).
b. RS02 radiated susceptibility, magnetic induction fields (figure 5).
c. RS03 radiated susceptibility, electric field, 14 kHz to 10 GHz.
d. CE03 conducted emissions, power and interconnecting leads, 1.5 MHz to 65 MHz (figure 3).
e. CS06 conducted susceptibility, spikes, power leads (figure 5).
f. CS02 conducted susceptibility, power and interconnecting leads, 0.05 to 400 MHz.

3.15 Weight. The weight of each, air conditioner, fully charged, shall not be more than as specified in table VII. Accessories specified in 3.17 shall not be included in the weight specified.

TABLE VII. Weight.

Type	Size	Maximum Weight (lb)
I	C	280
II	B	290

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3.16 Treatment and painting. All parts shall be cleaned, treated, and painted in accordance with MIL-T-704, type F or G, as applicable, in accordance with the top assemblies. Unless otherwise specified (see 6.2), the top coat shall be camouflage green 383, color 34094, in accordance with MIL-C-46168.

3.17 Accessories. When specified (see 6.2), accessories shall be provided with the air conditioners as specified herein:

3.17.1 Blast damper assembly and covers. A blast damper assembly (TA13222E9905), fresh-air cover (1322E9904), and special air conditioner cover (13226E1362), shall be provided for type I, size C, air conditioners. The blast damper shall permit full fresh-air flow when in the open position and shall not close with external-over pressure of 3 ±0.5 inches with pressure exceeding this value from above, either side, or from the back (condenser face). The damper shall close and remain closed until manually reset. The damper shall withstand a pressure of not less than 3.4 psig for a duration of not less than 0.6 seconds. The damper shall not leak, rupture, or be permanently deformed as a result of the 3.4-psig blast.

3.17.2 Electromagnetic protection assembly. An electromagnetic protection (EMP) assembly shall be provided when type II, size B, air conditioners are produced for systems with nuclear hardening requirements.

3.17.3 Block off assembly. A block off assembly for use with type I, size C, air conditioners shall be provided to cover the opening when the control box is removed for remote operation (see 3.10.1).

3.18 Identification marking. The air conditioner, components, and accessories shall be marked for identification in accordance with MIL-STD-130. Identification plates shall conform to MIL-P-15024 and MIL-P-15024/6.

3.19 Instruction plates. The air conditioner shall be equipped with instruction plates, labels, or diagrams, including warnings and cautions, describing procedures to be followed in operating or servicing the unit. Instruction plates shall conform to MIL-P-514.

3.20 Workmanship.

3.20.1 Metal fabrication. Metal used in fabrication shall be free from kinks and sharp bends. The straightening of material shall be done by methods that will not cause injury to the material. Corners shall be square and true. All bends shall be made by controlled means to ensure uniformity of size and shape. Precaution shall be taken to avoid overheating materials. Heated steel shall be allowed to cool slowly. External surfaces shall be free of burrs, sharp edges and corners, except when sharp edges or corners are required or where they are not detrimental to safety. Shop-fabricated components shall be assembled in steel jigs or fixtures and joined while held in position to minimize distortion.

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3.20.2 Bolted connections. Boltholes shall be accurately formed and shall have the burrs removed. Washers or lockwashers shall be provided where necessary. Matching thread areas securing bolts conforming to SAE J429 or capscrews shall be of sufficient strength to withstand the tensile strength of the bolt. All fasteners shall be correctly torqued and shall have full thread engagement.

3.20.3 Riveted connections. Rivets shall fill the holes completely. The upset rivet heads shall be full, neatly made, concentric with the rivet holes, and in full contact with the surface of the member, and shall be in accordance with SAE J492.

3.20.4. Welding. The surfaces of parts to be welded shall be free from oxide, scale, paint, grease, oil or any other foreign matter. Immediately before welding, oxide shall be removed by an abrading or a chemical process. Grease and oil shall be removed by wiping with a solvent such as acetone. All surfaces shall be dried after cleaning. Welds shall transmit stress without permanent deformation or failure when the parts connected by welds are subjected to proof and service loadings. Complete and uniform penetration and fusion of the metals shall be obtained on all welds. All welded parts shall be free from cracks and other imperfections that may reduce the effectiveness of the part. Work shall be positioned for flat welding whenever possible. Tack welds which will be incorporated in the final weld shall meet the same quality specifications as the final weld. All weld procedures shall be qualified in accordance with the applicable AWS or ASME codes as follows:

AWS D1.1, Structural Welding Code, Steel, Section 5, Qualification.
AWS D1.2, Structural Welding Code, Aluminum, Section 5, Qualification.
ASME Boiler and Pressure Vessel Code, Section IX, Welding Qualifications.

3.20.4.1 Welders. Before assigning any welder or welding operator welding work,the contractor shall obtain certification that the welder or welding operator has passed qualification tests as prescribed for the type of welding to be performed by that particular welder or welding operator and that such qualification is effective as defined by the particular code:

AWS D1.1 Structural Welding Code - Steel.
AWS D1.2 Structural Welding Code - Aluminum.
ASME Boiler and Pressure Vessel Code, Section IX, Welding and Brazing Qualifications.

Contractors who only make horizontal welds need not qualify welders for "all position welding". Certification shall be made available for review by the contracting officer or designated representative. In the event of poor welds, the Government reserves the right to require retesting of any welder or welding operator.

3.20.5 Brazing. Brazing shall be in accordance with MIL-B-7883. During the brazing operation a small amount of nonoxidizing gas shall be continuously bled through the refrigerant lines to prevent internal oxide scaling. During the

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brazing operation care shall be exercised to protect all components from deformation and damage. The quantity of brazing alloy and flux used shall be no more than that required to make the joint. Flux shall not be present in the refrigerant system and heavy beads and fillets of braze material shall be avoided.

3.20.6 Interchangeability. All parts having the same part number shall be functionally and dimensionally interchangeable. Interchangeable parts are defined as two or more parts possessing such functional and physical characteristics as to be equivalent in performance and durability and capable of being exchanged one for the other without alteration of the parts themselves or adjoining parts, except for adjustment, and without selection for fit or performance.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract of purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Disassembly inspection. Failure of any test or examination of the first article shall be cause for disassembly of the first article, in the presence of a Government representative, to the extent necessary to determine the cause of the failure. Each disassembled part shall be examined in detail for compliance with this specification and drawings referenced with regard to materials, dimensions, tolerances, and workmanship. Parts not complying with such requirements shall be rejected and shall be cause for rejecting the first article.

4.1.3 Component and material inspection. The contractor is responsible for insuring that all components and materials are manufactured, examined, and tested in accordance with referenced standards, specifications, and drawings, as applicable.

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4.2 Classification of inspections. The inspection requirements are classified as follows:

a. First article inspection (see 4.3 and 6.3).
b. Quality conformance inspection (see 4.4).
c. Inspection of packaging (see 4.6 and 6.7).

4.3 First article inspection.

4.3.1 First article examination. The first article air conditioners shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection and cause for performing the inspection specified in 4.1.2 as applicable.

4.3.2 First article tests. The first article air conditioners shall be subjected to the test marked "X" in columns 2 through 6 of table VIII in the sequence listed. Failure of any test shall be cause for rejection and performing the inspection specified in 4.1.2

4.4 Quality conformance inspection.

4.4.1 Examination. Each air conditioner shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.4.2 Tests. Each air conditioner shall be subjected to the tests marked "X" in column 7 of table VIII in the sequence listed. Failure of any test shall be cause for rejection.

4.5 Examination procedure.

4.5.1 Examination. The air conditioner shall be examined for the following defects:

Major

101. Parts or components missing (see 3.1).
102. Parts (or components) not in accordance with QAP requirements as shown in the drawings (see 3.1).
103. Nonconformance to the drawings (see 3.1.1).
104. Materials not as specified (see 3.4).
105. Materials not resistant to corrosion and deterioration, or treated to be resistant to corrosion and deterioration for the applicable storage and operating environments (see 3.4.1).
106. Dissimilar metals as defined in MIL-STD-889 are not effectively insulated from each other (see 3.4.2).
107. Contractor does not have documentation available for identification of material, material finishes, or treatment (see 3.4.3).
108. Used, rebuilt or remanufactured components, pieces or parts incorporated in the air conditioner (see 3.4.4).
109. Human factors engineering requirements not as specified (see 3.6).

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110. Electrical circuitry not protected from the effects of moisture and fungus growth (see 3.8.5).
111. Air conditioners not fully charged with the type of refrigerant specified (see 3.9).
112. Material Safety Data Sheets not as specified (see 3.9.6).
113. Controls not as specified (see 3.10 thru 3.10.5). Each incorrect control shall constitute one defect.
114. Installation of wiring not as specified (see 3.12.1).
115. Electromagnetic interference suppression components and network not identical to approved model 2/ (see 3.14).
116. Weight of air conditioners not as specified (see 3.15).
117. Treatment and painting not as specified or painting defects such as blisters, thin spots, lumps, scratches, runs or sags (see 3.16).
118. Color not as specified (see 3.16).
119. Accessories not provided as specified (see 3.17 thru 3.17.3). Each missing accessary shall constitute one defect.
120. Identification marking not as specified (see 3.18).
121. Instruction plates not provided as specified (3.19).
122. Workmanship not as specified (see 3.20).
123. Metal fabrication not as specified (see 3.20.1).
124. Jigs and fixtures not used as specified (see 3.20.1).
125. Washers or lockwashers missing (see 3.20.2).
126. Bolted connections not as specified (see 3.20.2).
127. Riveted connections not as specified (see 3.20.3).
128. Welding not as specified (see 3.20.4).
129. Welders not certified as specified (see 3.20.4.1).
130. Brazing not as specified (see 3.20.5).
131. Interchangeability not as specified (3.20.6).

4.5.2 Tests.

4.5.2.1 Test schedule.

2/ The approved model shall be the approved first article #1.

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TABLE VIII. Test schedule. 1/

Test req	First article test number 1	2	3	4	5	Qual conf	Test	Test paragraph	Requirement paragraph
1	2	3	4	5	6	7	8	9	10
1	X	X	X	X	X	X	Refrigeration system pressure.	4.5.2.3.1	3.9.5
2	X	X	X	X	X	X	Examination	4.5.1	3.1, 3.4, 3.6, 3.8.5, 3.9, 3.10, 3.12.1, 3.15, 3.16, 3.17, 3.18
3	X	X	X	X	X	X	Controls.	4.5.2.4	3.10.2, 3.10.3, 3.10.4, 3.9.3, 3.9.4, 3.12.2, 3.12.3
4	X	-	-			-	Refrigerant moisture content.	4.5.2.5	3.9.1
5	X	-	-			-	Electromagnetic interference.	4.5.2.20	3.14
6	X	X	X	X	X	X	Evaporator compartment air leakage.	4.5.2.7	3.11
7	X	-	-			-	Cooling capacity and airflow.	4.5.2.6	3.5.2, 3.5.3, 3.5.4, 3.5.6
8	X	-	-			-	Heating capacity.	4.5.2.8	3.5.5.1
9	X	-	-			-	Evaporator coil Frost Switch	4.5.2.9	3.10.5
10	-	X	-			-	High temperature operation, cooling.	4.5.2.10	3.5.1.1, 3.8.2
11	-	X	-			-	High temperature storage.	4.5.2.11	3.8.1
12	-	X	-				Low temperature operation cooling/ heating.	4.5.2.12	3.5.1.2, 3.5.5.2, 3.8.2
13	-	X	-			-	Low temperature storage.	4.5.2.13	3.8.1
14	-	X	-			-	Rain.	4.5.2.14	3.8.4
15	X	-	-				Humidity.	4.5.2.15	3.8.3
16	-	X	-			-	Salt fog.	4.5.2.16	3.8.3
17	-	X	-			-	Vibration.	4.5.2.17	3.13
18	X	X	-			X	Run-in.	4.5.2.18	3.5
19	X	X	-			X	Refrigeration system leak.	4.5.2.3.2	3.9.5

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TABLE VII. Test schedule. 1/

Test req	First article test number 1	2	3	4	5	Qual conf	Test	Test paragraph	Requirement paragraph
1	2	3	4	5	6	7	8	9	10
20	-	-	X	X	X	-	Reliability.	4.5.2.19	3.3.1
21	-	-	X	X	X	-	Maintenance evaluation.	4.5.2.21	3.3.2
22	X	X	-			-	Safety.	4.5.2.22	3.7
23	-	X	-				Variable voltage and frequency.	4.5.2.23	3.12.4, 3.12.2
24	X	-	-			-	Blast damper. 2/	4.5.2.24	3.17.1
25	X						Human factors engineering.	4.5.2.25	3.6
26	X	-	-			-	Noise.	4.5.2.26	3.6.1

1/ Perform inspection marked "X".
2/ Test applies only when blast damper is specified for some or all of the items being procured.

4.5.2.2 Conditions. Unless otherwise specified herein, tests shall be conducted using 208V, 60-Hz input electrical supply. Unless otherwise specified herein, the frequency tolerance for all tests shall be ±1 percent. Unless otherwise specified herein, no braces, insulating or sealing materials, tiedown devices, or other items that are not inherently a part of the operating air conditioner shall be employed. Unless otherwise specified herein, no alignment of components, addition of refrigerant, or other servicing shall be performed on the preproduction or production air conditioners after testing of the air conditioner has begun. Components subject to atmospheric corrosion shall not be treated with any treatment such as grease or oil to meet test requirements.

4.5.2.3 Refrigerant system pressure and leakage.

4.5.2.3.1 Refrigerant system pressure. Charge the refrigerant system to 425 psig +5 psig or -0 psig with dry nitrogen containing refrigerant gas detectable with a halogen-sensitive electronic leak detector (General Electric type H-2 or equal). Reduce the pressure 300 ±5 psig. Calibrate the leak detector for the type of gas and percentage of mixture employed and set the detector to indicate a leakage rate of 0.1 ounce per year corrected to the percentage of the mixture. Check the entire refrigerant system for leaks. One or more of the following shall constitute failure of this test:

a. Permanent deformation of any component in the refrigerant system.
b. Any detectable refrigerant leak.

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4.5.2.3.2 Refrigerant system leakage. Test for leakage at room ambient temperature (70 °F or above) with a halogen-sensitive electronic leak detector (General Electric type H-2 or equal). Set the detector to indicate a leakage rate of 0.1 ounce per year. Check the entire refrigerant system for leaks. A detectable leak shall constitute failure of this test.

4.5.2.4 Controls.

4.5.2.4.1 Mode selector switch and temperature selection control. With power connected to the air conditioner, position the mode selector switch and temperature selection control in the following positions:

a. OFF position. Determine that the air conditioner components have been deenergized as specified in 3.10.2 and the compressor crankcase heater is energized.
b. VENT position. Determine that the evaporator fan motor is energized and operating, and that the compressor motor, condenser fan motor (if separate), and the heaters are off.
c. COOL position-temperature selection control at coolest temperature position. Subject the thermistor sensor to air having a temperature of 70 +15 or -5 °F. Determine that the evaporator and condenser fan motor(s) and compressor motor have ramped up to full speed and are operating and the heaters are off. Determine that there is refrigerant flow to the expansion valve.
d. COOL position-temperature selection control at warmest temperature position. Subject the thermistor sensor to air having a temperature not exceeding 80 °F. Determine that the compressor motor is off and the evaporator fan motor continues to operate. Determine that the pressure equalization valve has opened and that the low and high side pressures have equalized.
e. HEAT position-temperature selection control at warmest temperature position. Subject the thermistor sensor to air having a temperature of 70 +5 or -15 °F. Determine that the evaporator fan motor is energized and operating and the compressor motor and condenser fan motor (if separate) are off. Determine that the heaters are energized and operating. Then position the temperature selection control at minimum temperature position and determine that the heaters are off.

Nonconformance to 3.9.2, 3.10.2, or 3.10.3 shall constitute failure of this test.

4.5.2.4.2 Heater high-limit cut-out switch. Operate the air conditioner in heat mode at the warmest temperature setting. After 15 minutes (min) of operation, deenergize the evaporator fan to increase the temperature within the evaporator compartment. Determine that the heaters are deenergized. Nonconformance to 3.10.4 shall constitute failure of this test.

4.5.2.4.3 High-pressure cutout switch. The high pressure cutout switch shall be tested as follows:

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a. First article. With the air conditioner operating in the cooling mode, at the coolest temperature setting, gradually block off the condenser airflow until the compressor is stopped by action of the high-pressure cutout switch. Determine that the TEMPERATURE/PRESSURE fault indicator light turns on. Record the pressure at which the compressor is stopped. Cutout pressure not as specified in table V or failure of the fault indicator to light shall constitute failure of this test.
b. Quality conformance.

(1) Prior to installation in the air conditioner, bench test the high-pressure cutout switch. Using dry nitrogen, increase the pressure to the switch until it opens (cuts out). Record the pressure at which the switch opens. Repeat this test procedure two more times. Cutout pressures not as specified in table V shall constitute failure of this test.
(2) After completion of the bench test, the high-pressure cutout switch shall be tested in the air conditioner with the air conditioner operating in the cooling mode. With the compressor operating, press the high pressure switch reset button. The compressor shall cease to operate. Failure of the compressor to cease operating at the time the reset button is pressed and the TEMPERATURE/PRESSURE fault light to illuminate shall constitute failure of this test.

4.5.2.4.4 Low pressure cutout switch. Electrically connect a continuity meter to the pins of the low pressure switch. The meter shall show a closed circuit. Connect a low pressure gage to the suction service valve. Bleed the gas from the unit while constantly monitoring the suction pressure and continuity meter. The continuity meter shall show an open circuit when the suction pressure reduces to the low pressure specified in table V. Subsequent to this test, charge the unit with refrigerant only. The following shall constitute failure of this test:

a. The continuity meter indicating an open circuit prior to charge bleedoff.
b. Failure of the continuity meter to show an open circuit when the suction pressure reduces to the low pressure specified in table V.
c. Failure of the temperature/pressure fault light to illuminate.

4.5.2.4.5 Overcurrent fault indicator. Connect an adjustable 10 kW, 208V, 3 Phase resistive load bank to the motor controller output leads. Set the mode selector switch to COOL, the temperature selection control to coolest temperature and the load bank at no electrical current flow. With the air conditioner operating, slowly increase the electrical current flow to the load bank until the motor controller overcurrent protection circuit is activated and the air conditioner ceases operation. Determine that OVERCURRENT fault indicator is lighted. Record the current at which the OVERCURRENT protection actuates. Failure of the overcurrent fault indicator to light at the time air conditioner ceases to operate constitutes a failure of this test.

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4.5.2.4.6 Over/under voltage fault indicator. Set the mode selector switch to the COOL position and the temperature selection control to coolest temperature. With the air conditioner operating, slowly reduce the input voltage until the under voltage protection circuit is activated and the air conditioner ceases operation. Determine the OVER/UNDER VOLTAGE fault indicator is illuminated. Record voltage at which the protection circuit actuated. Reset, and restart the air conditioner. Slowly increase the input voltage until the over voltage protection circuit activates and the air conditioner ceases operation. Determine that the OVER/UNDER VOLTAGE fault indicator is lighted. Record the input voltage at which the protection circuit actuated. Failure of the fault indicator to light when the air conditioner shuts off constitutes a failure of this test.

4.5.2.4.7 Hot motor controller fault indicator. Install a thermocouple on the motor controller cooling surface between the cooling fins. Set the mode selector switch on COOL and the temperature selection control to coolest temperature. With the air conditioner operating, slowly restrict the air flowing across the cooling fins until the overtemperature protection circuit activates and the air conditioner ceases operation. Determine that the HOT MOTOR CONTROLLER fault indicator is lighted. Record the motor controller surface temperature at which the protection circuit actuated. Failure of the fault indicator to light when the air conditioner shuts off constitutes a failure of this test.

4.5.2.5 Refrigerant moisture content. Operate the air conditioner for not less than 1 hour in the cooling mode, in an ambient temperature of not less than 95 °F. Shut down the air conditioner, and after a period of 1 hour, withdraw a sample of approximately 1 pound of refrigerant from the low-pressure side and determine the moisture content by one of the methods specified in BB-F-1421 or by the titrimetic method using the Karl Fischer reagent. A moisture content of more than 25 ppm shall constitute failure of this test.

4.5.2.6 Cooling capacity and airflow. Test the air conditioner for cooling capacity at the capacity rating conditions specified in 3.5.2 and the evaporator fan for airflow at the conditions specified in 3.5.4.

4.5.2.6.1 Cooling capacity. Capacity tests shall be conducted using the instruments, apparatus, and procedures specified in ASHRAE Standard 16 and as specified herein. Capacity tests shall be conducted with type I air conditioners located with the front face (evaporator side) flush with the inside room surface of the wall between the inside and outside room. Type II air conditioners shall be located, with respect to the inside and outside room for the capacity tests, as shown on figure 1. Prior to the capacity tests, thermocouples shall be installed at the thermal expansion valve bulb, liquid line at the expansion valve inlet, and hot gas compressor discharge line. After installation, the thermocouples shall be completely insulated to prevent ambient temperatures from affecting the temperatures obtained from them. In addition to the data required by the applicable ASHRAE standards, temperatures obtained from the above thermocouples shall be recorded at the same times that the ASHRAE data are recorded. These temperatures shall be used, in the event of capacity test failures, to troubleshoot the refrigeration system.

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4.5.2.6.2 Airflow. Airflow tests shall be conducted using the instruments, apparatus, and procedures specified in ASHRAE Standard 37 or ASHRAE Standard 51/ AMCA Standard 210 (nozzles only for air flow measurement).

4.5.2.6.3 Basis of failure. Nonconformance to 3.5.2, 3.5.3, 3.5.4, 3.5.6, or 4.5.2.2 shall constitute failure of this test.

4.5.2.7 Evaporator compartment air leakage. Close the fresh-air damper if necessary (see 3.11). Box the entire front face (evaporator side) of the air conditioner so that the evaporator is at least 2 inches from the box surface facing it and air is free to circulate over the entire evaporator side (from the top of the evaporator discharge opening to the base of the air conditioner). Seal the box to the air conditioner so that air does not leak between the air conditioner and the box. Attach a duct equipped with an air-circulating and measuring device to the CBR or fresh-air opening (see 3.11) in the air conditioner. Seal the connection between the air conditioner and duct to prevent external air leakage. Seal condensate drains. Other sealing that is not a part of the air conditioners shall not be permitted except as required for instrumentation lines. By means of the air-circulating device, subject the evaporator compartment to static pressure of 1.0-inch wg and maintain the pressure for not less than 5 minute while measuring the airflow leakage. Airflow leakage shall be measured using the instruments, apparatus, and procedures specified in the ASHRAE Standard 37 or ASHRAE Standard 51 (nozzles only) or AMCA Standard 210. Nonconformance to 3.11 shall constitute failure of this test.

4.5.2.8 Heating capacity. The test shall be conducted in an ambient temperature between 75 and 80 °F. Prior to recording any data, operate the air conditioner for a period of not less than 2 hours with the mode selector switch turned to the HEAT position and the temperature selection control set for warmest temperature. Calculate the heating capacity using the following equation:

$$QH = 3.414 \times W$$

where QH = Heating capacity in Btu/hr
W = Power input (watts)

Measure the power input (W) to the air conditioner. Nonconformance to 3.5.5.1 or 3.5.6 shall constitute failure of this test.

4.5.2.9 Evaporator coil frost switch. Insert a thin thermocouple approximately 2 1/2 inch into the evaporator coil fins from the top, with the thermocouple as close as possible to the coil frost switch capillary tube. This shall be used with an electronic temperature indicator to monitor the coil (and switch capillary) temperature. With the unit operating in the cooling mode at shop ambient temperature, allow a controlled amount of air to bypass the evaporator coil. This can be accomplished by partially removing the evaporator cover (keeping the condenser compartment covered). By controlling the amount of evaporator bypass air, allow the coil temperature to slowly decrease until the coil frost switch opens and shuts off the compressor. Record the temperature at which the switch opens. Monitor the switch as the coil temperature rises and

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record the temperature at which the coil frost switch again makes contact. Perform this procedure 3 times. The average of the readings shall be within ±1 °F of the specified switch set points. Nonconformance with 3.10.5 shall constitute failure of this test.

4.5.2.10 High-temperature operation cooling. Subject the air conditioner to a temperature between 120 and 123 °F for a period of 4 hours. After completion of the exposure period, operate the air conditioner in the cooling mode, at the coolest temperature setting, for a period of not less than 1 hour in an ambient temperature between 120 and 123 °F with air between 120 and 123 °F entering the evaporator and condenser. Inability of the air conditioner to start and operate or nonconformance to 3.5.1.1 or 3.5.3 shall constitute failure of this test.

4.5.2.11 High temperature storage. Subject the entire air conditioner to a temperature of 160 °F for a period of 4 hours. At the conclusion of the exposure period, remove the air conditioner from the test chamber and examine for deterioration, corrosion, or damage. After completion of the examination, allow the air conditioner to soak at ambient temperature between 65 and 85 °F for a period of not less than 1 hour but not more than 3 hours. After completion of the soaking period, operate the air conditioner in the cooling mode, at the coolest temperature setting, for a period of not less than 1 hour. Inability of the air conditioner to start, operate, or any evidence of deterioration, corrosion, damage, such as rupture of tubing, damage to control components, binding of motors, fans or other mechanical parts, deterioration of gaskets, damage to insulation, breakdown of adhesives, permanent setting of gaskets or packing, or loss of refrigerant shall constitute failure of this test.

4.5.2.12 Low temperature operation. Install sheet metal ducts on the evaporator-return-air-inlet and evaporator-air-discharge openings. Each duct shall be not less than 12 inches long and of the same cross sectional dimensions as the respective openings. Equip each duct with a multipoint thermocouple grid located in the duct 6 inches from the front of the air conditioner. The thermocouples of each grid shall be connected in parallel to obtain an average temperature value.

4.5.2.12.1 Low temperature operation cooling.

4.5.2.12.1.1 Type I, size C and type II, size B. Operate the air conditioner continuously for 12 hours in the cooling mode in an ambient temperature of 55 ±5 °F, with air at +70 °F, and 60 °F wet bulb ±3 °F, entering the evaporator. Immediately following the 12-hour operation, inspect the evaporator coil. Any evidence of ice or frost on the finned area of the evaporator coil, or failure of the air conditioner to operate and provide cooling for 12 hours shall constitute failure of this test.

4.5.2.12.1.2 Type II, size B. Place the air conditioner in a test chamber, installed for in-the-wall operation, with the air conditioner front exposed to a temperature of 70 ±5 °F and the rear (external portion) exposed to 0 ±2 °F. With power connected to the air conditioner, maintain the temperatures for a 10 hour soak period. Then, with the temperature selection control set for coolest

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temperature, start and operate the air conditioner in the cool mode for a period of not less than 1 hour with air at 70 ±5 °F entering the evaporator, and the air at 0 ±2 °F entering the condenser. One or more of the following shall constitute failure of this test:

a. Inability of the air conditioner to start and operate in the cooling mode.
b. Temperature differential between the average entering air temperature and average leaving evaporator air temperature less than 15 °F when operating in the cooling mode after stabilization of conditions has occurred.

4.5.2.12.2 Low temperature operation, heating. After completion of 4.5.2.12.1, subject the entire air conditioner to -50 ±5 °F for a period of not less than 4 hours. After completion of the 4-hour soak period, start and operate the air conditioner in the heat mode of operation, at the warmest temperature setting, for a period of not less than 1 hour with air at -50 ±5 °F entering the evaporator. Determine that heaters are energized and operating. Turn the air conditioner off for a period of 15 minutes. While maintaining the -50 ±5 °F temperature, restart and operate the air conditioner in the heat mode of operation for a period of not less than 15 minutes. Inability of the air conditioner to start, operate or restart in the heat mode shall constitute failure of this test.

4.5.2.13 Low temperature storage. Subject the air conditioners to a temperature of -60 °F for a continuous period of 12 hours. At the conclusion of the exposure period, remove the air conditioners from the test chamber and examine for deterioration and damage. After completion of the examination, allow the air conditioners to soak at ambient temperature between 65 and 85 °F for a period of not less than 1 hour. After completion of the soaking period, operate the air conditioners in cooling mode at the coolest temperature setting, for a period of not less than 1 hour. Any evidence of deterioration or damage or failure of the air conditioner to start and operate shall constitute failure of this test.

4.5.2.14 Rain. Subject the air conditioner to test method 506, procedure 1 of MIL-STD-810 as specified herein. The air conditioner shall be installed on an enclosure to simulate a shelter application. Type I air conditioner evaporator sections shall be flush with the opening of an enclosure. Type II air conditioner sections shall protrude through the opening of an enclosure with the enclosure partitions over the evaporator/condenser partition of the housing. Seal the opening to prevent any water from entering the enclosure. Rainfall shall be produced by means of water spray and shall be measured with U.S. Weather Bureau type gages. Nozzles shall emit droplets having a diameter of not less than 1.5 millimeters. Water supply temperature shall be between 50 and 70 °F and shall contain fluorescein dye for ease in determining the locations and extent of leakage. After each rain test, remove all panels, access doors, and component covers from the air conditioner as required to examine for water penetration. After the examination, operate the air conditioner for 15 minutes in the cooling mode and for 15 minutes in the heating mode.

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4.5.2.14.1 Free rain. With the air conditioners not operating, the condenser covers rolled up, and fresh air doors closed, subject each of the three exposed sides and top of the air conditioners to a simulated rainfall of 4 ±1-inches per hour for 30 minutes each, for a total test duration of not less than 2 hours. Direction of rainfall against each of the three sides shall be varied throughout the test from vertical to 45 degrees from vertical.

4.5.2.14.2 Blowing rain. With the air conditioners operating in the cooling mode, at the coolest temperature setting, and the fresh air doors closed, subject each of the three exposed sides to a simulated rainfall of 1 ±0.5-inch per hour combined with a horizontal wind of 40 ±5 miles per hour for a period of 1 hour. Total test duration shall be not less than 3 hours.

4.5.2.14.3 Basis of failure. One or more of the following shall constitute failure of the rain test:

a. Evidence of water leaking into the evaporator enclosure.
b. Malfunctioning of components or an electrical short during operation.
c. Nonconformance to 3.8.4.

4.5.2.15 Humidity. Subject the air conditioner to test method 507, procedure I, of MIL-STD-810 except the maximum temperature shall be 155 °F. Do not operate the air conditioner during exposure and do not expose the air conditioner to extreme temperature prior to test. Prior to this test and after the exposure period, operate the air conditioner in accordance with 4.5.2.4.1 to determine proper functioning of the controls. Nonconformance to 3.8.3 or 4.5.2.4.1 shall constitute a failure of this test.

4.5.2.16 Salt fog. Subject the air conditioner to test method 509, procedure I, of MIL-STD-810. Do not operate the air conditioner during exposure. Prior to this test and after the exposure period, operate the air conditioner in accordance with 4.5.2.4.1 to determine proper functioning of the controls. Nonconformance to 3.8.3 or 4.5.2.4.1 shall constitute failure of this test.

4.5.2.17 Vibration. Conduct the vibration test as specified in MIL-STD-810, method 514.2, category F, procedure VIII, table 514.2-VI, figure 514.2-6, curve V (see 6.8). The test time shall be 60 minutes total per plane. The air conditioners shall not be operated during this test and shall be mounted as follows:

a. Type I air conditioners shall be mounted by bolts using the mounting receptacles provided in the bottom and above the condenser-fan discharge grille.
b. Type II air conditioners shall be mounted by bolts using the mounting by bolts using the mounting plates provided in the bottom.
c. Auxiliary braces, straps, or other tiedown devices, other than those specified herein, shall not be employed to secure the air conditioners to the test apparatus. No blocking or tiedown of internal components other than those that are part of the air conditioner shall be employed. Removable panels and access doors shall be in place during

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all testing. With the air conditioner mounted, there shall be no resonance inherent in the test apparatus at the frequencies and acceleration levels specified for this test. During the tests, the base of the air conditioner shall remain in a horizontal plane. Mount accelerometers on the test apparatus immediately adjacent to the four corners of the base of the air conditioner. The four accelerometers shall be connected into an electric or electronic network which will average the accelerometers' output signals into a single indicated g level of the magnitude specified for this test. After completion of each test axis, conduct the controls test specified in 4.5.2.4.1 and the refrigeration system leak test specified in 4.5.2.3.2.

4.5.17.1 Basis of failure. Nonconformance to 3.13, 4.5.2.3.2, or 4.5.2.4.1 shall constitute failure of this test.

4.5.2.18 Run-in.

4.5.2.18.1 Run-in performance. Install ducts on the evaporator-return-air openings, evaporator-air-discharge opening, and condenser air inlet opening of each air conditioner. Each duct shall be not less than 12-inches long and of the same cross-sectional dimensions as the respective openings. Equip each duct with a thermocouple grid to measure average entering and leaving air temperatures to the evaporator and entering air to the condenser. Measure the entering evaporator-air wet-bulb temperature by psychrometer device located in the test chamber. Operate the air conditioners for not less than 1 hour in the cooling mode, at the coolest temperature setting, while maintaining the supply voltage to the air conditioner within ±5 percent. The air temperature entering the evaporator return and condenser inlet shall be as specified in table IX. After stabilization, three sets of the data specified in table X shall be recorded and averaged during the last 30 minutes of operation (every 10 minutes). After completion of the above, operate the air conditioner in an ambient temperature between 100 and 120 °F for a total period of 48 hours in the cooling mode, at the coolest temperature setting, in cyclic operation of 55 minutes "ON" and 5 minutes "OFF". After completion of the 48-hour operation, continue operating the air conditioner in the cooling mode, at the coolest temperature setting, for not less than an additional hour while maintaining the supply voltage to the air conditioner within ±5 percent and temperature of the air entering the evaporator return and condenser inlet as specified in table IX. After stabilization, three sets of the data specified in table X shall be recorded and averaged during the last 30 minutes of operation (every 10 minutes). Operate the air conditioner in heat mode, at the warmest temperature setting, in an ambient temperature between 65 and 85 °F for 30 minutes. During the last 10 minutes of operation, record the power input (W) to the air conditioner. After completion of the 30-minute operating period, conduct the refrigerant system leak test as specified in 4.5.2.3.2.

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TABLE IX. Temperature conditions for run-in performance.

Location	Temperature
Evaporator return.	Dry bulb - 90 ±3 °F.
Evaporator return.	Wet bulb - dry-coil condition.
Condenser inlet.	Dry bulb - 110 ±3 °F.
Condenser inlet.	Wet bulb - dry-coil condition.

4.5.2.18.2 Run-in performance curves. Not less than 25 units shall be used to establish acceptance curves (see 6.6). When less than 25 units are produced, all units will be used to establish acceptance curves. Data obtained in accordance with 4.5.2.18.1 shall be used by the contractor to select one high and one low unit that then shall be capacity-tested at condition "A" of 3.5.2. If any high- or low-limit air conditioner fails to meet minimum requirements of 3.5.2, it shall be replaced by the next low or high unit and that unit tested. Operate the acceptable high and low air conditioner continuously in the cooling mode for not less than 1 hour under each of the conditions specified in table XI. After stabilization, four sets of data specified in table X shall be recorded and averaged for each of the conditions in table XI during the last 30 minutes (every 10 minutes). Establish high and low limits of acceptance for production units by plotting the following curves from data specified in table X:

a. Average evaporator-entering dry-bulb temperature versus average temperature differential between evaporator inlet and outlet air temperatures for each of the condenser-entering temperatures recorded.
b. Average evaporator-entering dry-bulb temperature versus kW power input for each of the condenser-entering temperatures.
c. Average evaporator-entering dry-bulb temperature versus suction pressure for each of the condenser-entering temperatures.
d. Average evaporator-entering dry-bulb temperature versus discharge pressure for each of the condenser-entering temperatures.

TABLE X. Data required for run-in performance curves.

Dry-bulb temperature of air entering the evaporator, °F.
Wet-bulb temperature of air entering the evaporator, °F.
Dry-bulb temperature of air leaving the evaporator, °F.
Dry-bulb temperature of air entering the condenser, °F.
Condensing pressure, psig.
Suction pressure, psig.
Power consumption of the air conditioner, W.

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TABLE XI. Temperature conditions for run-in performance curves.

Location	Temperature			
	Condition I	Condition II	Condition III	Condition IV
Evaporator return	87 ±0.5 °F	93 ±0 5 °F	87 ±0.5 °F	93 ±0.5 °F
Condenser inlet	107 ±0.5 °F	113 ±0.5 °F	113 ±0.5 °F	107 ±0.5 °F
Evaporator return and condenser inlet: Dry coil condition, all conditions.				

4.5.2.18.3 Failure criteria. One or more of the following shall constitute failure of this test:

a. Temperature differential between the average air dry-bulb temperature entering the evaporator return and the average air dry-bulb temperature leaving the evaporator before and after the 48-hour test phase in 4.5.2.18.1 not within the curves developed as specified in 4.5.2.18.2.
b. Average condensing pressure recorded before and after the 48-hour test phase in 4.5.2.18.1 not within the curves developed as specified in 4.5.2.18.2.
c. Average suction pressure recorded before and after the 48-hour test phase in 4.5.2.18.1 not within the curves developed as specified in 4.5.2.18.2.
d. Average power consumption on cooling recorded before and after the 48-hour test phase in 4.5.2.18.1 not within the curves developed as specified in 4.5.2.18.2.
e. Total power consumption (W) on heating (maximum heat) in excess of that specified in table III, column 3.
f. Failure of any component or fracture or breakdown of any material.
g. Evidence of any detectable refrigerant leak.
h. Need for any servicing of the air conditioner during operation.

4.5.2.19 Reliability. Using the upper and lower test MTBFs specified in 3.3.1, the air conditioners shall be tested as specified herein with accept and reject criteria in accordance with MIL-HDBK-781, test plan XIV. In addition, the occurrence of more than 2 failures of the same type, or attributable to the same component shall constitute failure of this test. Each air conditioner tested must accumulate at least one third of the total test hours.

4.5.2.19.1 Test facility. Install the air conditioners in a test chamber consisting of two rooms. One room shall function as an indoor room and shall maintain the evaporator return-air temperature within ±5 °F of the temperatures specified in figure 2. The second room shall function as an outdoor room and shall maintain the condenser return air temperature within ±5 °F of the temperatures specified in figure 2. These temperatures shall be maintained on a 24-hr cyclic basis throughout the reliability test, as shown on figure 2.

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4.5.2.19.2 Operation. The mode selector switch and temperature selection control shall be operated manually or automatically. If automatic means are used, they shall not interrupt the control circuit but shall physically operate the air conditioner control components. The controls shall be positioned as shown in table XII. The air conditioner shall be operated in accordance with the standardized test profile shown in figure 2 for the duration specified in 4.5.2.19. Vertical units shall receive 208v, 3 phase input with one each receiving 50, 60, and 400 Hz. For horizontal units, one shall receive 230 v, single phase, 50 Hz input; one shall receive 208v, 3 phase, 60 Hz input; and the remaining unit shall receive 208v, 3 phase, 400 Hz input.

TABLE XII. Control positions for reliability test.

Mode	Time	Control	Control Position
Heating	0030 to 0130	Mode Switch	Heat
Heating	0030 to 0100	Thermostat*	Warmest
Heating	0100 to 0130	Thermostat*	Coolest
Vent	0130 to 0400	Mode Switch	Vent
Vent	0130 to 0400	Thermostat*	Warmest
Cooling	0400 to 2100	Mode Switch	Cool
Cooling	0400 to 2100	Thermostat*	Coolest
Cooling	2100 to 2400	Mode Switch	Cool
Cooling	2100 to 2400	Thermostat*	Warmest
Off	0000 to 0030	Mode Switch	Off

*Temperature selection control.

4.5.2.19.3 Data. The following shall be monitored and recorded throughout the test as required to determine that normal air conditioner operation is being maintained and to identify problems in event of failure or malfunction:

a. Voltage.
b. Current.
c. Frequency.
d. Power factor.
e. Refrigerant pressures.
   (1) Suction.
   (2) Discharge.

f. Refrigerant temperatures.
   (1) Compressor discharge.
   (2) Liquid line.
   (3) Suction line (at compressor).

g. Air temperatures.
   (1) Evaporator return (dry bulb).
   (2) Evaporator discharge (dry bulb).
   (3) Condenser return (dry bulb).

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h. Abnormal operation (visual and audio observation).
i. Sight-glass observation (clear or bubbles).
j. Identification of unit.
k. Initials of technician.
l. All failures.
m. Time of day of failure.
n. Action taken if failure occurs.

4.5.2.19.4 Failure criteria. A failure is defined as any malfunction for which corrective action cannot be deferred until the next scheduled routine maintenance check, provided corrective maintenance action can be performed at the organizational level with available tools, or which cannot be corrected by the operator crew with available tools within 30 minutes. Corrective action is not deferrable if the malfunction causes or may cause inability to commence operation of the unit, cessation of operation of the unit, or degradation of performance capability as specified herein. Malfunctions will not be counted as "failures" for reliability purposes if detected and corrected during initial inspection or if they may be deferred to final inspection (except for those that cause test termination). Malfunctions that result from operational or maintenance errors will not be counted. A malfunction shall be considered to have occurred if the following happens:

a. Temperature difference across the evaporator coil is less than as specified in table XIII when the air conditioner is in the cooling mode.
b. Current draw is greater than as specified in table XIII when the air conditioner is in the cooling mode with the temperature selection control in the coolest position.
c. Current draw is greater than as specified in table XIII when the air conditioner is in the heating mode with the temperature selection control in the warmest position.
d. Current draw does not decrease when the air conditioner is in the heating mode and the temperature selection control is changed from the warmest position to the coolest position.
e. Temperature at the evaporator discharge does not increase when the air conditioner is turned to the heating mode with the temperature selection control set in the warmest position.
f. The sight-glass is not clear when the air conditioner is in the cooling mode.
g. The discharge pressure is greater than specified in table II (condensing pressure column under 3.5.2) when the air conditioner is in the cooling mode.
h. The suction pressure is lower than specified in table II (suction pressure column under 3.5.2) when the air conditioner is in the cooling mode.
i. Any abnormal operation occurs as determined by looking at or listening to the unit.

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TABLE XIII. Minimum temperature drop and maximum current draw. 1/

Unit	Input Power	Temperature difference °F (minimum)	Current draw Amps (max)
Type I C	208V, 3 phase, 60 Hz	20.0	21
Type II B	230V, single phase, 60 Hz	20.0	35
	208V, 3 phase, 60 Hz	20.0	21

1/ For all 3-phase air conditioners, amperage reading shall be the average of the three phases.

4.5.2.20 Electromagnetic interference. Using the test plan specified in MIL-STD-462, the air conditioner shall be tested in accordance with MIL-STD-461 using test methods:

a. REO2 radiated emissions, electric field, 0.15 MHz to 1000 MHz, broadband and narrow band.
b. RSO2 radiated susceptibility, magnetic induction fields.
c. RSO3 radiated susceptibility, electric field, 14 KHz to 10 GHz.
d. CEO3 conducted emissions, power and interconnecting leads, 1.5 MHz to 65 MHz.
e. CSO2 conducted susceptibility, power and interconnecting leads, 0.05 to 400 MHz.
f. CSO6 conducted susceptibility, spikes, power leads.

4.5.2.20.1 Test conditions. The above tests will performed under the following steady state and transient conditions, operating with power supplied at 208V/3Ph/50, 60 and 400 Hz and 230V/1Ph/60Hz (type II only).

Steady-state conditions. Controls of the air conditioner shall be set for the following modes of operation at steady-state conditions:

STEADY-STATE TEST MODES

Mode Control setting	Temperature selection setting
Cool	Coolest
Cool	Warmest
Heat	Warmest
Heat	Coolest

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Transient conditions. Transient durations shall be measured when the air conditioner controls are switched between the following modes:

TRANSIENT TEST MODES

Mode	Mode Control setting	Temperature selection setting
Off to Vent	Vent	
Vent to cool	Cool	Coolest to warmest
Cool	Cool	Warmest to coolest
Heat	Heat	Warmest to coolest
	Heat	Coolest to warmest

4.5.2.20.2 Basis of failure. One or more of the following shall constitute failure of the electromagnetic interference test:

a. Steady-state emissions in excess of those shown on figures 3 and 4.
b. Transient emissions above the limits specified on figures 3 and 4 in excess of 1-second duration.
c. Loss of control functions, damage to air conditioner, erratic operation or false indication of faults during susceptibility tests.

4.5.2.21 Maintenance evaluation. The maintenance ratio shall be computed as specified in 3.3.2 for all maintenance actions required during or as a result of first article testing. Nonconformance to 3.3.2 shall constitute failure of this test.

4.5.2.22 Safety. Evaluation shall be accomplished throughout testing to determine conformance to MIL-STD-882. Electrical leakage from the air conditioner casing to ground shall be measured in all operating modes, including the OFF position when connected to a power source. Nonconformance to 3.7 shall constitute failure of this test.

4.5.2.23 Variable voltage and frequency test.

4.5.2.23.1 Cooling mode.

4.5.2.23.1.1 Start-up. At room ambient or at rated temperature conditions start the air conditioner at each of the voltage and frequency conditions specified in table VI in order to monitor the in-rush current to the air conditioner. An oscillograph will be used to record the in-rush current during start-up. The existence of in-rush current values to the air conditioner greater than specified in table XIII shall constitute failure of this test.

4.5.2.23.1.2 Steady state operation. Subject the entire air conditioner to a temperature of 120 ±3 °F for a period of 1 hour. After completion of the exposure period, start and operate the air conditioner in the cooling mode, at the coolest temperature setting, for a period of not less than 1 hour in an ambient

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temperature of 120 ±3 °F, with air at 120 ±3 °F entering the evaporator, at each of the voltage and frequency conditions specified in table VI. Record data either continuously or at not greater than 15 minute intervals during each test. Data shall include:

a. Air temperatures: Dry bulb in and out of evaporator and condenser; wet bulb into the evaporator.
b. Refrigerant pressures: High and low sides of system.
c. Power characteristics: Steady State voltage, current, watts and instantaneous current (using an oscillograph) during the first five seconds of operation.

Failure of air conditioner to start or operate at all the conditions shall constitute failure of this test.

4.5.2.23.2 Heating mode.

4.5.2.23.2.1 Start-up. At ambient or at rated conditions start the air conditioner at each of the voltage and frequency conditions specified in table VI in order to monitor the in-rush current to the air conditioner. An oscillograph will be used to record the in-rush current during start-up. The existence of in-rush current values to the air conditioner greater than specified for table VI shall constitute failure of this test.

4.5.2.23.2.2 Steady state operation. Subject the entire air conditioner to a temperature of -50 ±5 °F for a period of 1 hour. After completion of the exposure period start and operate the air conditioner in the heating mode, at the warmest temperature setting, for a period of not less than 1 hour in an ambient temperature of -50 ±5 °F with air at -50 ±5 °F entering the evaporator at each of the voltage and frequency conditions specified in table VI. Record data either continuously or at not greater than 15 minute intervals during each test. Data shall include:

a. Air temperatures: Dry bulb in and out of evaporator.
b. Power characteristics: Steady State voltage, current, watts and instantaneous current (using an oscillograph) during the first five seconds of operation.

Failure of the air conditioner to start or operate at all the conditions shall constitute failure of this test.

4.5.2.24 Blast damper. The blast dampers shall be tested on an operating air conditioner with the evaporator side at normal atmospheric pressure. The back, sides and the top of the air conditioner shall then be subjected to three series of blast pressures. Each series shall consist of a sufficient quantity of blast pressures to ascertain that the dampers remain open at 2.5 inches water gage (wg), close at 3.6 wg, and can withstand a minimum of 3.4 psig.

4.5.2.24.1 Basis of failure. One or more of the following shall constitute failure of the blast damper test:

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a. Failure to remain open at 2.5 inches wg.
b. Failure to close at 3.6 inches wg.
c. Evidence of permanent deformation, rupture, or leakage.

4.5.2.25 Human factors engineering. The air conditioner shall be evaluated to determine that the human factors criteria have been met. Nonconformance to 3.6 shall constitute failure of this test.

4.5.2.26 Noise level test. To determine the noise level of the air conditioner at the evaporator and condenser sections, place a microphone on an arc 36 inches from the center of the evaporator side and 60 inches from the center of the condenser side in the same horizontal plane. Record the noise level while the air conditioner is in normal operation with the compressor running. In each section, readings shall be taken directly in line with the geometric center of the evaporator and condenser discharge openings of the air conditioner; at the evaporator section, 30 degrees to the left and 30 degrees to the right of the center. At the condenser section, the spread of the microphone shall be 45 degrees vice 30 degrees. Nonconformance to 3.6.1 shall constitute failure of this test.

4.6 Inspection of packaging.

4.6.1 First article pack inspection. The first article pack shall be examined for the defects specified in 4.6.3. Presence of one or more defect shall be cause for rejection.

4.6.2 Quality conformance inspection of packaging.

4.6.2.1 Unit of product. For the purpose of inspection, a complete pack prepared for shipment shall be considered a unit of product.

4.6.2.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.6.3 Examination. Samples selected in accordance with 4.6.2.2 shall be examined for the following defects. Acceptable quality level (AQL) shall be 1.0 percent defective.

132. Preservation methods not as specified for level A or C (see 5.2).
133. Louvers not closed and canvas covers, if supplied, not snapped in place as specified for level A or C (see 5.2.1.1.1 and 5.2.2.1).
134. Technical publications not preserved and placed with the air conditioner as specified for level A or C (see 5.2.1.1.2 and 5.2.2.1).
135. Accessories not cushioned, placed in bag or box, and positioned as specified for level A or C (see 5.2.1.1.3 and 5.2.2.1).
136. Air conditioners not protected with a shroud as specified for level C (see 5.2.2.2).
137. Cushioning not of the type, size, or positioned as specified as level A or B (see 5.2.1.1.3).
138. Strapping not as specified for level A or B (see 5.3.1.1 and 5.3.2.1).

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139. Clearance between air conditioner and inner faces of box not as specified for level A or B (see 5.3.1.1 and 5.3.2.1).
140. Air conditioners requiring bolting to the base of the box not bolted as specified for level A or B (see 5.3.1.1 and 5.3.2.1).
141. Air conditioners not packed in accordance with the reference document for level C packing (see 5.3.3).
142. Palletization, when appropriate, not as specified for all levels of packaging (see 5.4).
143. Marking missing, illegible, incorrect, or incomplete for level A, B, or C (see 5.5).

5. PACKAGING

5.1 First article pack. The contractor shall furnish a first article pack for examination within the time frame specified (see 6.2) to prove, prior to starting production packaging that the applied preservation, packing and marking comply with the packaging requirements of this specification. Examinations shall be those specified in section 4 and shall be subject to surveillance and approval by the Government (see 6.7). The first article pack may be accomplished using either the first article model air conditioner or a production air conditioner. If the first article model air conditioner is used, any preservation and packing shall be removed by the contractor, when requested by the Government at no expense to the Government to facilitate comparison between the first-article-model air conditioner and production air conditioners.

5.2 Preservation. Preservation shall be level A or C as specified (see 6.2).

5.2.1 Level A.

5.2.1.1 General.

5.2.1.1.1 Canvas covers and louvers. Prior to applying the method II barrier material around the air conditioner, as required in 5.2.1.2, all louvers shall be placed in the closed position. Fresh air door control chains, if so equipped, shall be secured in place. Canvas covers, if supplied, shall be snapped in place.

5.2.1.1.2 Technical publications. Technical publications shall be preserved in accordance with MIL-P-116, method IC-1 or IC-3 and placed with the air conditioner within the method II barrier specified in 5.2.1.2.

5.2.1.1.3 Accessories.

a. Block-off assemblies for type I air conditioners, when specified, shall be placed in a sack conforming to PPP-S-30, type II, which shall be secured shut with staples. For the type II air conditioners, parts of the vibration mounts shall be placed with the above accessories. Electrical connectors shall be wrapped with cushioning material conforming to PPP-C-843 prior to placement in the sack with other accessories. The consolidated accessories shall be secured with tape to the air

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conditioner in a location that will not interfere with any cushioning or blocking; however, on type I, size C air conditioners, the accessories shall be placed behind the junction box in the condenser compartment.

b. Blast damper assemblies, and fresh air covers, when specified, shall be individually packed in boxes conforming to PPP-B-636, type CF, class domestic, variety SW, weight appropriate for the weight of contents. Prior to placement in the boxes, each accessory shall be wrapped with cushioning material conforming to PPP-C-843. Special air conditioning covers, when specified, shall be placed in a polyethylene bag having a film thickness of not less than 2.5 mils. The bag shall be secured shut with staples or by heat sealing.

5.2.1.2 Air conditioners: type I, size C and type II, size B. Air conditioners shall be preserved in accordance with MIL-P-116, method IIa. Air conditioners shall be secured to the bottom of boxes specified in 5.3.1.1 with four bolts which pass through the barrier material.

5.2.2 Level C.

5.2.2.1 General. All louvers shall be closed, and canvas covers shall be snapped in place. Technical publications and accessory items (block-off assemblies, cable connectors, vibration mounts, etc.) shall be preserved and secured to the air conditioner as specified for level A.

5.2.2.2 Air conditioners. Air conditioners shall be covered with a polyethylene film shroud of not less than 3 mils in thickness. The shroud shall be secured to the air conditioner with tape.

5.3 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.3.1 Level A.

5.3.1.1 Air conditioners: type I, size C, and type II, size B. Air conditioners with all included accessories preserved as specified in 5.2, shall be packed in a box conforming to PPP-B-601, overseas type. The box size shall be such that not less than 1-inch clearance shall exist between faces of the air conditioner and the inner walls of the box. The air conditioner shall be secured to the bottom of the box with four bolts, using the boltholes provided in the bottom of the air conditoner. The box shall be modified with skids in accordance with the box specification except that the skids shall be located so that the bolts securing the air conditioner pass through the skids. Boltheads shall be countersunk into the bottom of the skids. Strapping shall conform to QQ-S-781, class I, type I or IV, finish B, size as applicable.

5.3.2 Level B.

5.3.2.1 Air conditioners: type I, size C and type II, size B. Air conditioners and all accessories preserved as specified in 5.2 shall be packed as specified in 5.3.1.1 except the boxes shall be packed domestic type and required

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strapping shall be finish A or the air conditioners shall be packed in a container as shown in figure 6. The box shall be closed and sealed with tape in accordance with the appendix to the box specification.

5.3.3 Level C. Air conditioners, preserved as specified in 5.2, shall be packed in accordance with the packing requirements of MIL-STD-1190.

5.4 Palletization. When a multiple quantity of air conditioners is being shipped to the same destination, packed units for all levels shall be palletized. Only those air conditioners whose packed size will permit two, three, or four units on a 40 x 48 inch pallet will be palletized. Pallets shall conform to NN-P-71, 40 x 48 inches, type optional. Air conditioners shall be secured to the pallet with strapping conforming to QQ-S-781, class 1, type I or IV, 0.75 x 0.028 inches minimum. For level A palletized loads strapping shall be finish B. Strapping shall be arranged so that each air conditioner on the pallet is restrained in all directions.

5.5 Marking. In addition to any special marking specified in the contract or purchase order (see 6.2), marking shall be in accordance with MIL-STD-129 and the special marking specified in 5.5.1.

5.5.1 Special marking. Each shipping container shall be marked to indicate that the air conditioners shall remain in the operating position during shipment and storage. Special markings shall consist of arrows and the words "THIS END UP." Special marking shall be placed within the upper third of each side of the shipping container in black letters not less than 3 inches high.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The air conditioners covered by this specification are intended for use in cooling, dehumidifying, heating, filtering, and circulating air in portable shelters or vans to meet equipment and personnel requirements. The air conditioner may be installed indoors or directly exposed to the weather.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1, 2.2 and 6.8).
c. Definitive specification part number of air conditioner required (see 6.10).
d. When a block-off assembly is required for type I, size C air conditioners (see 3.17).
e. When an EMP assembly is required for type II, size B air conditioners (see 3.17).

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f. When a blast damper assembly, with fresh air cover and special air conditioner cover, is required for type I, size C air conditioners (see 3.17).
g. When a first article is required for inspection and approval, and the number of units required (see 3.2 and 6.3).
h. Time frame required for submission of first article (see 3.2 and 6.3).
i. When the Government will conduct any or all of the first article model examination and tests. When the Government will conduct some but not all of the first article examination and tests, the contracting officer should specify which examination and tests will be conducted by the Government and which examination and tests shall be conducted by the contractor (see 3.2 and 6.3).
j. Color required when other than camouflage green (see 3.16).
k. Time frame required for submission of the first article pack (see 5.1).
l. Level of preservation and packing required (see 5.2 and 5.3).
m. Any special marking required (see 5.5).

6.3 First article. When a first article inspection is required, the units should be preproduction models. The first article should consist of five units. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, tests, reports, and approval of the first article test results, and disposition of the first article. Any changes or deviations of production units from the approved first article will be subject to the approval of the contracting officer. Approval of the first article will not relieve the contractor of his obligation to furnish air conditioners conforming to this specification.

6.4 Quality assurance provisions (QAP). The contracting officer should require the contractor to maintain records of all QAP inspections. A suggested paragraph is as follows:

"The contractor shall maintain complete records of all examinations and test performed to verify the requirements of classified QAP characteristics. The records shall include as a minimum, lot size, sample size, drawing requirement, actual measurement, number and type of deficiencies found, quantity approved, quantity rejected, and corrective action taken when applicable."

6.5 Definitions.

6.5.1 Quality assurance provisions (QAP). A QAP is a contractual requirement that supplements section 4 of the specification. QAP indicates the minimum requirements that must be inspected on the product drawings to verify the design objectives of the product and assure interchangeability of repair parts.

6.5.2 Definitive military specification part number. The military specification part number corresponds to the type and size of the air conditioner covered by this specification.

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6.6 Technical manuals. The requirement for technical manuals should be considered when this specification is applied on a contract. If technical manuals are required, military specifications and standards that have been cleared and listed in DOD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL) must be listed on a separate Contract Data Requirements List (DD Form 1423), which is included as an exhibit to the contract. The technical manuals must be acquired under separate contract line item in the contract. In addition, the contracting officer should require that copies of the performance curves specified in the run-in test (see 4.5.2.18.2) be furnished to the engineering agency within 30 days after development.

6.6.1 Material Safety Data Sheets. Contracting officer should identify those activities requiring copies of completed Material Safety Data Sheets prepared in accordance with FED-STD-313. The pertinent government mailing addresses for submission of data are listed in Appendix B of FED-STD-313 (see 3.9.6). As a minimum, data should be submitted to the procuring activity Safety Officer.

6.7 First article pack. Any changes or deviations of production packs from the approved first article pack will be subject to the approval of the contracting officer. Approval of the first article pack will not relieve the contractor of his obligation to preserve, pack, and mark the air conditioners in accordance with the specification.

6.8 Vibration.

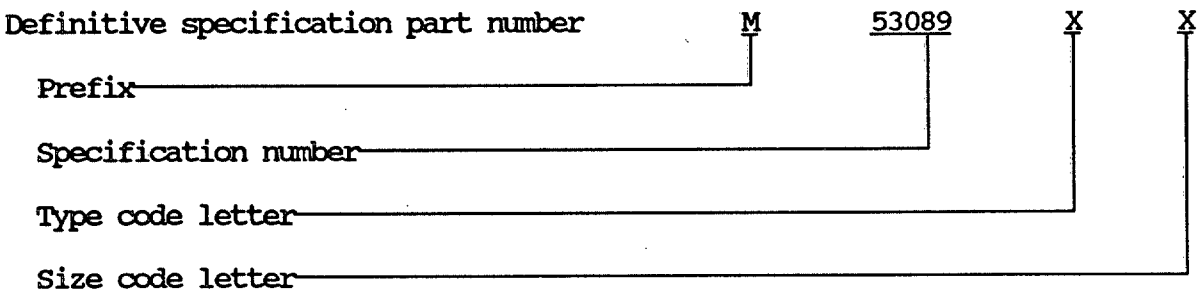
a. These air conditioners have been previously qualified to MIL-STD-810, revision C, under criteria specified in section 3 and 4 of this specification. Subsequent notices or revisions should not be made applicable to a procurement contract.
b. The contracting officer should include the following in procurement documents: "Revisions other than those listed below are not applicable. MIL-STD-810, Revision C."

6.9 Superseding data. This specification supercedes MIL-A-52767 in part as follows:

FROM: MIL-A-52767	TO: MIL-A-53089
Type I-Vertical	Type I-Vertical
Size Class	Size
C 1-208V, 3-phase, 50/60 Hz	C - 208V, 3-phase, 50/60/400 Hz
2-208V, 3-phase, 400 Hz	
Type II-Horizontal	Type II-Horizontal
Size Class	Size
B 1-230V, single phase, 50/60 Hz	B - 230V, single phase, 50/60 Hz
2-208V, 3-phase, 50/60 Hz	- 208V, 3-phase, 50/60/400 Hz
3-208V, 3-phase, 400 Hz	

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6.10 Part or Identifying system. The part or identifying number (PIN) of air conditioners covered by this specification shall be designed in the following form (see 6.2):



6.11 Subject term (key word) listing.

Air conditioner
Environmental control unit
Horizontal air conditioner
Motor controller
Multiple power input
Vertical air conditioner

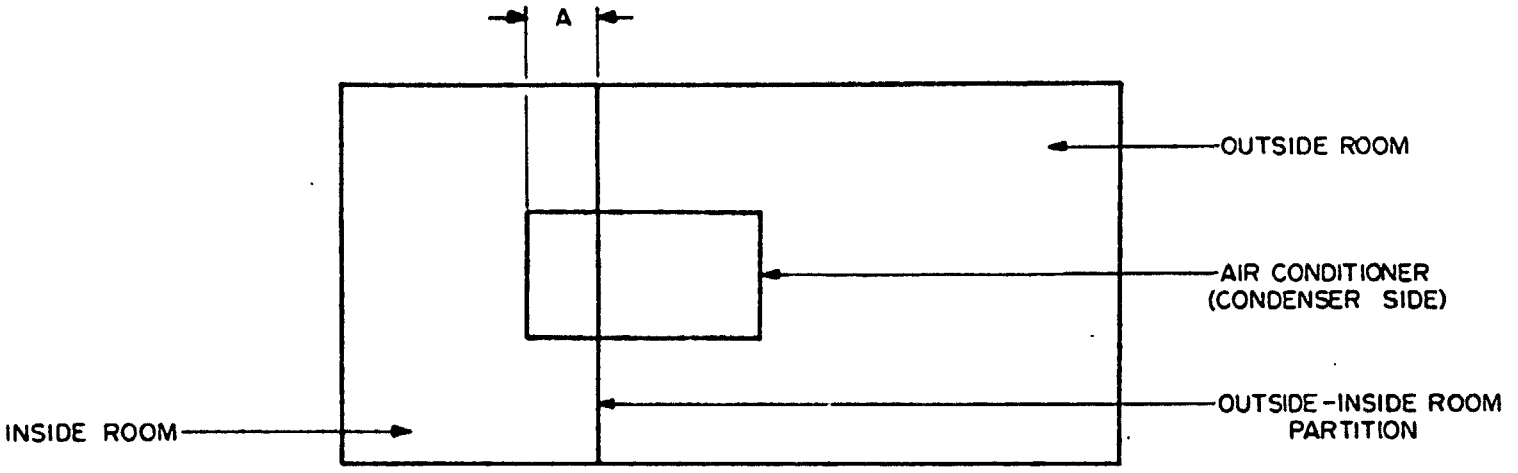
Custodians:
Army - ME
Navy - MC
Air Force - 99

Review activities:
Air Force - 82
DLA - GS

Preparing activity:
Army - ME

Project 4120-0319

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AIR CONDITIONER		"A" DIMENSION
TYPE	SIZE	INCHES
II	A	10.25 ± 0.5
II	B	11.50 ± 0.5
II	C	14.75 ± 0.5
II	D	15.75 ± 0.5

FIGURE 1. Type II air conditioner locations for capacity test

X-2440A

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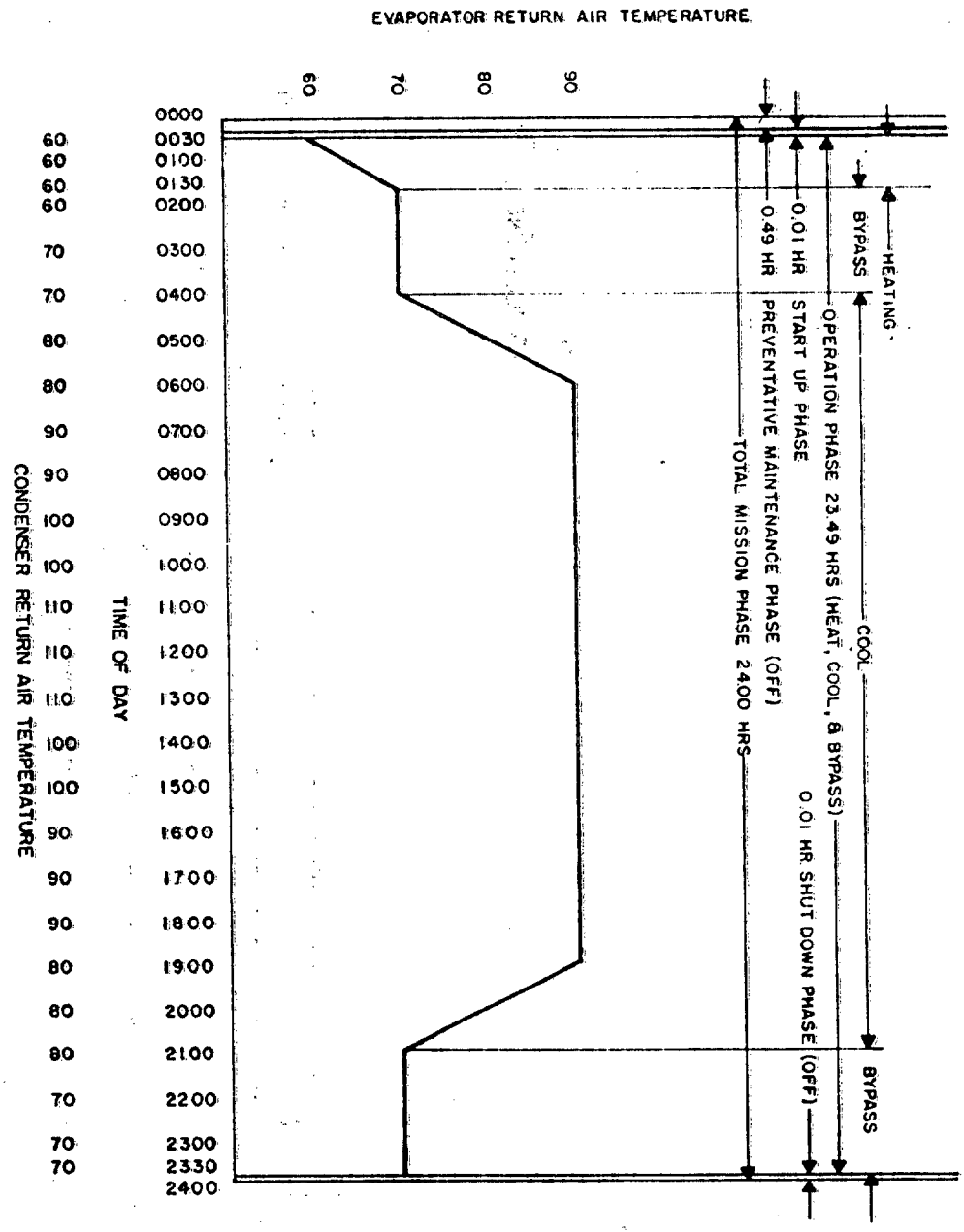


FIGURE 2. STANDARDIZED TEST PROFILE.

X-2441

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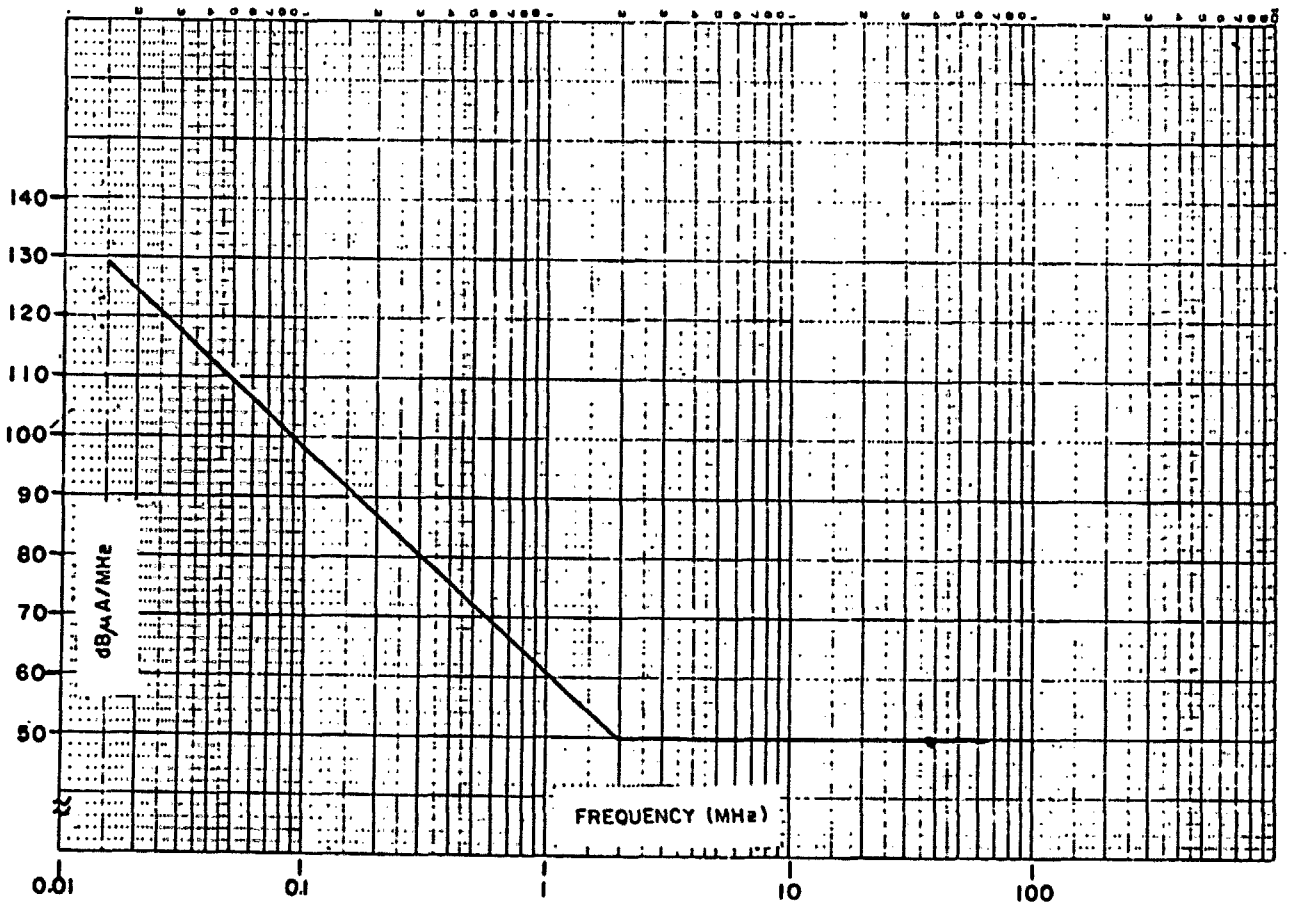


FIGURE 3. Limit for CE03 broadband emissions DC and interconnecting leads and AC leads.

X-3747A

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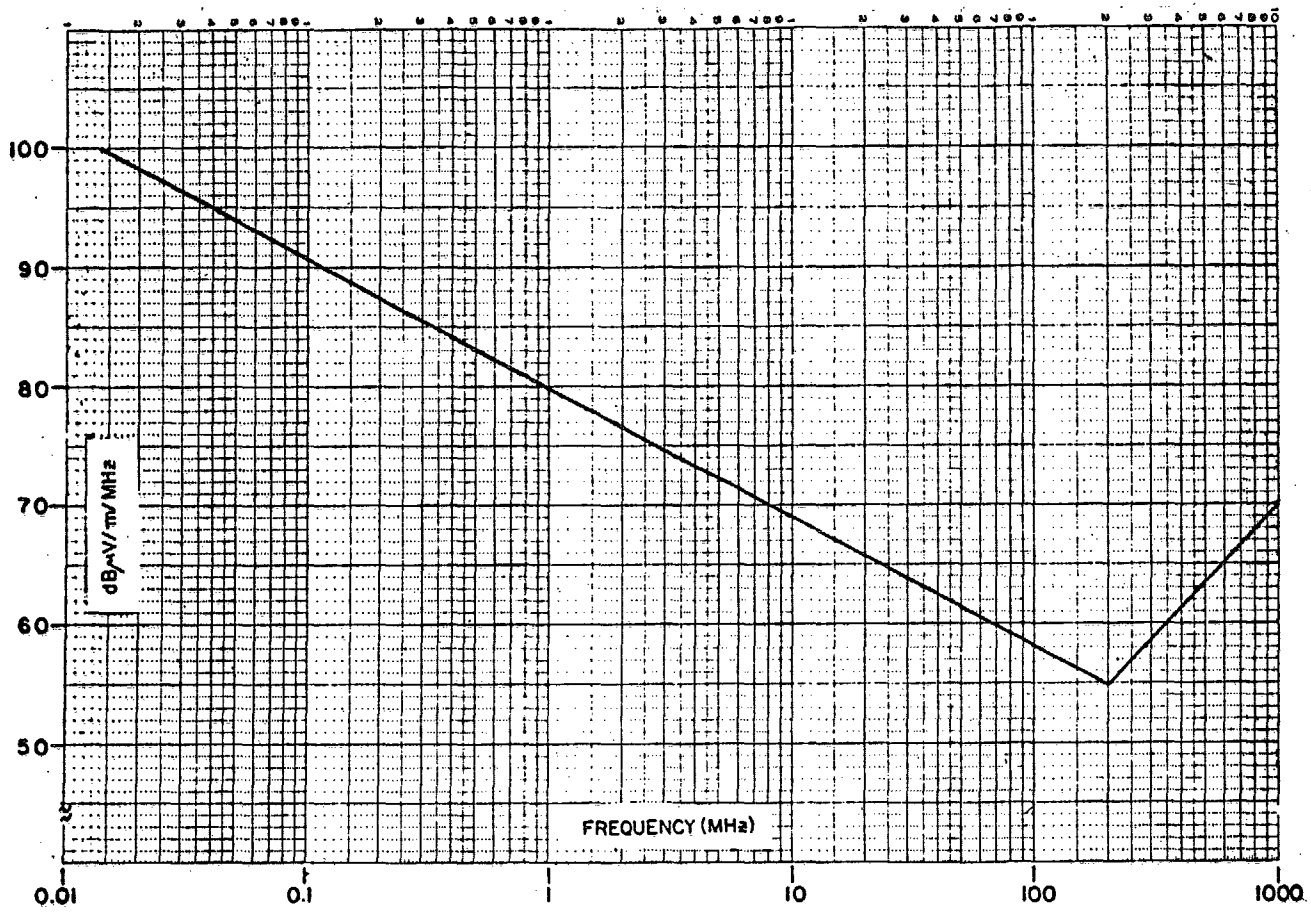
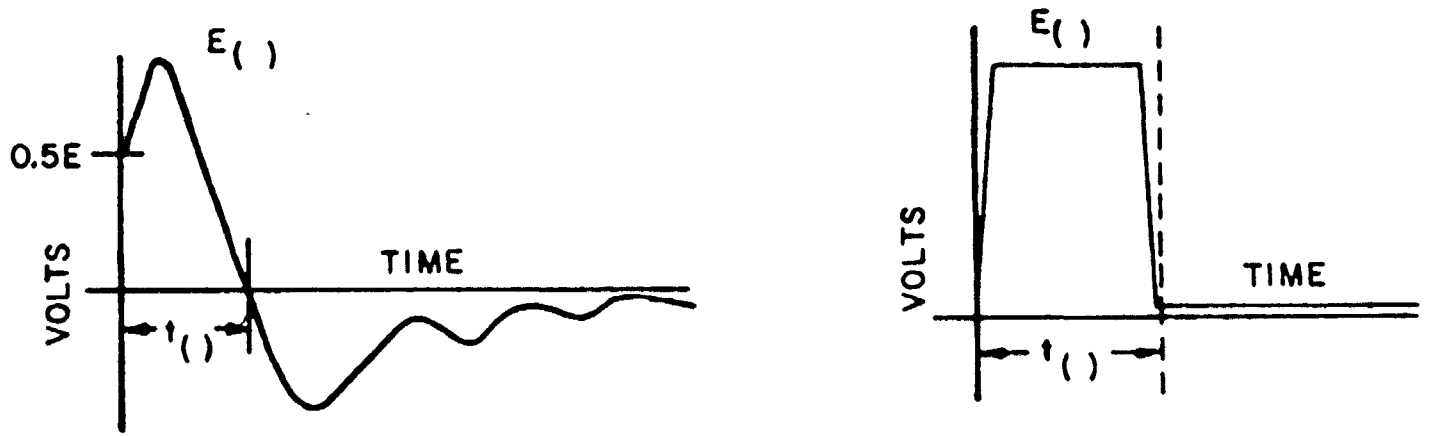


FIGURE 4. Limit for REO2 broadband emissions.

X-3748

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NOTE: THE TEST SAMPLE SHALL BE SUBJECTED TO THE SPIKE(S) WITH THE WAVEFORM SHOWN AND WITH THE SPECIFIED VOLTAGE(S) AND PULSEWIDTH(S).

FIGURE 5. Acceptable waveshapes for CS06 and RS02.

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NOTES

1. ALL DIMENSIONS ARE IN INCHES.
2. BOX MATERIAL SHALL BE WEATHER RESISTANT, TRIPLE-WALL CORRUGATED FIBERBOARD IN ACCORDANCE WITH PPP-B-640.
3. THE EXACT SIZE OF THE BASE AND COVER SHALL BE DETERMINED BY ACHIEVING A SNUG FIT OVER THE CUSHIONING AROUND THE AIR CONDITIONER AND NOISE ATTENUATOR
4. STRAPPING SHALL BE IN ACCORDANCE WITH QQ-S-781, CLASS 1, TYPE I OR IV, 3/4 X 0.015 OR EQUIVALENT.
5. ATTACH COVER TO BASE USING NAILS OR STAPLES.
6. TOP SEAMS SHALL BE SEALED WITH WATERPROOF TAPE.
7. PLYWOOD SHALL CONFORM TO PS 1, GRADE C-D INTERIOR WITH EXTERIOR GLUE; OR TO ANSI/HPMA HP 1983, GRADE 3-4, TYPE I.
8. SKIDS SHALL BE IN ACCORDANCE WITH PPP-B-601.

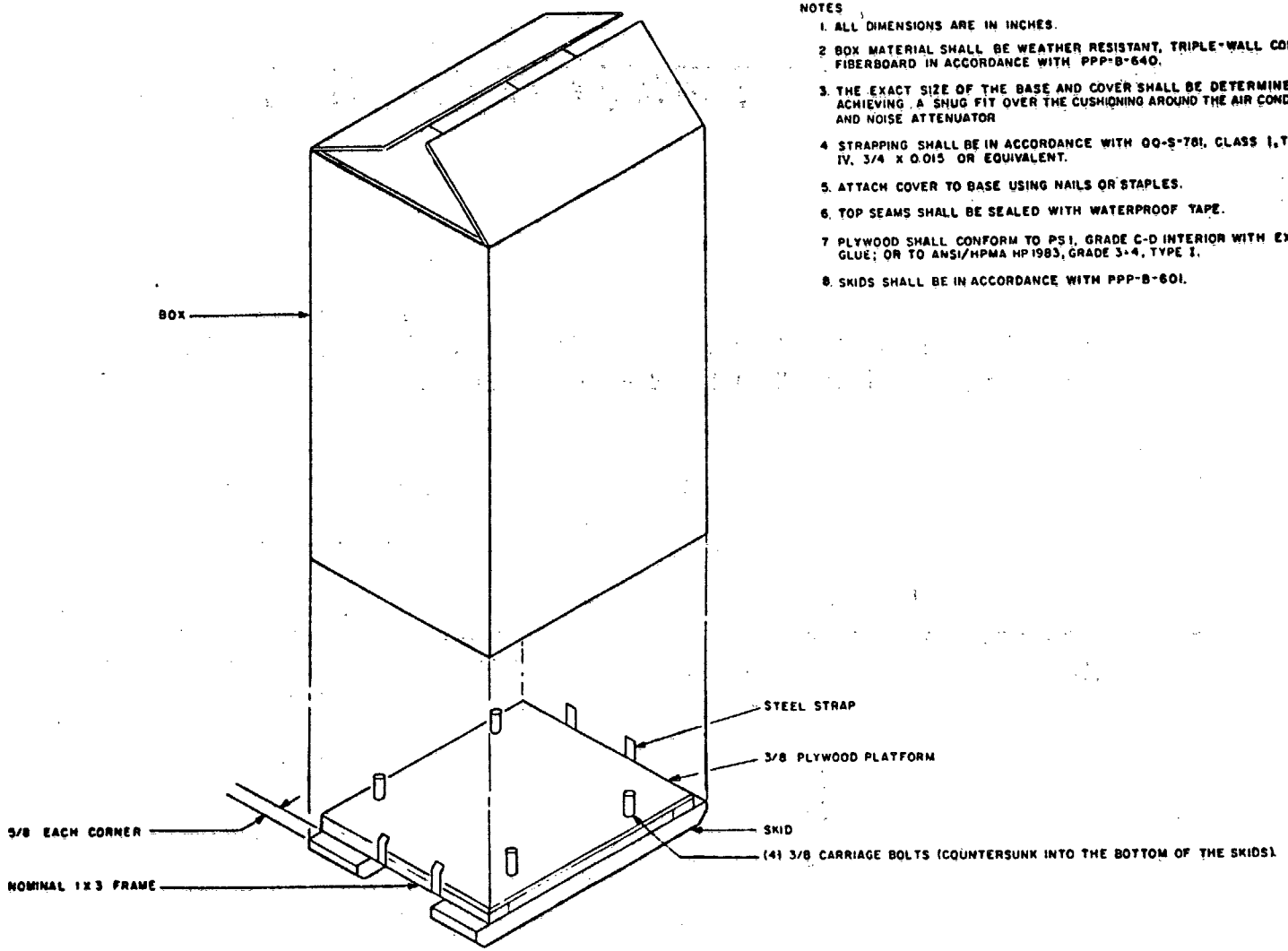


FIGURE 6. Shipping container Level B.

X-3597A

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-A-53089	Air Conditioners: Vertical and Horizontal, Compact, Multiple Power Input

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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